scReg-Eval: a fixed-panel audit of regulatory alignment in single-cell RNA foundation-model gene graphs

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ABSTRACT

Background. Claims that single-cell RNA foundation models encode gene regulation are difficult to audit because many reference networks share co-expression structure with the models' training data.
Methods. We introduce **scReg-Eval**, a fixed-panel audit protocol and software capsule that compares frozen pairwise foundation-model gene graphs with a sequence- and accessibility-derived regulatory-potential proxy while controlling co-expression, target peak count, gene length, GC content, RNA detection rate, transcription-factor out-degree, and target in-degree. The protocol-frozen panel contains 1,200 genes, including 446 in-range transcription factors (TFs). We evaluate thirteen pooled model/readout combinations across brain and paired peripheral blood mononuclear cell (PBMC) multiome data using two plus-one Monte Carlo randomization tests ($N = 999$ each): a gene-label permutation and a within-TF-row, degree-preserving target shuffle. Benjamini–Hochberg correction is applied within eight prespecified tissue-by-confound-specification-by-null families. The estimand is fixed-panel unusualness of alignment under named confounds and two named randomizations—not causal TF–target recovery and not model ranking.
Results. Protocol-pass is $0/13$ on the thirteen pooled rows. Dual-null "Support" under the primary full confound specification is seven of thirteen rows ($7/13$; six small positive alignments, ρ from 0.00425 to 0.01241, plus one negative Geneformer attention row in brain). That census is not regulatory recovery and is not a cross-family multiplicity correction.
The same seven rows appear only when degree covariates computed from the proxy are conditioned on; under the prespecified non-degree sensitivity dual-null Support is $0/13$ and three of the six positives change sign. Five of the six positive dual-null rows fail the protocol's own FM–co-expression concordance check (Spearman -0.18 to -0.02), while PBMC scGPT weakly passes (Spearman $+0.01$) yet still fails protocol-pass. The co-expression baseline, tested on the same edge set outside FM BH families (uncorrected p), is itself dual-null significant in both tissues, and the PBMC Geneformer-embedding alignment ($\rho = 0.00425$) is smaller than its own baseline ($\rho = 0.0070$). External ENCODE pan-biosample TF-occupancy edges give brain Spearman $\rho = 0.081$ and PBMC $\rho = 0.109$ (mean per-TF AP ≈ 0.51–0.53 among TFs with ≥ 5 positives), a failed occupancy check.
A supervised TF-disjoint probe on paired PBMC data gives mixed family-level results: UCE versus co-expression is supported by paired sign-flip testing ($q = 0.0325$), as is the random-init floor ($q = 0.0300$), whereas Geneformer embedding ($q = 0.5125$), Geneformer attention ($q = 0.65$), and scGPT ($q = 0.255$) are not; those contrasts mark relative over-correction of the baseline, not recovered regulation.
Conclusions. The audit finds weak, readout-specific, specification-dependent residual alignment in encoder and embedding readouts plus one opposite-signed attention row. It neither supports a global regulatory-capability claim for these scFM gene graphs nor licenses a field-level negative conclusion beyond this fixed-panel protocol instance.

Keywords: single-cell RNA-seq, foundation models, motif-accessibility proxy, co-expression confounding, randomization inference, benchmark audit

1 INTRODUCTION

Single-cell RNA foundation models (scFMs) are increasingly used as frozen representations for network inference and perturbation analysis (Theodoris et al., 2023; Cui et al., 2024). A high score against a curated or transcriptome-derived regulatory network, however, need not imply that a model encodes regulation beyond co-expression (Pratapa et al., 2020; Marbach et al., 2012; Nourisa et al., 2025). Both the reference and the model can inherit gene abundance, detectability, hubness, and co-variation, so agreement with a co-expression-laden reference can look like regulatory recovery when it is mostly shared structure. Standardized scFM task suites and living GRN benchmarks have begun to systematize evaluation (Qiu et al., 2025; Wu et al., 2025; Nourisa et al., 2025), but most still score expression-derived or literature GRNs rather than separating edge weights from expression statistics in the reference construction. Without that separation, it is hard to tell whether an scFM gene graph carries residual regulatory alignment after co-expression and simple graph-degree structure are controlled.

We address this problem as an audit-design question rather than as a model-ranking exercise. scReg-Eval freezes a transcription-factor (TF)–target panel, constructs an accessibility-and-motif regulatory-potential proxy whose *edge weights* are assembled without expression statistics, applies the same edge mask and confounds to every model, and evaluates the observed alignment against two structurally different randomization nulls. Panel membership and ATAC peak presence remain expression-adjacent design choices, so the proxy is not claimed to be RNA-structure-independent in every upstream decision—only that its scored edges are not co-expression weights. This design does not make the proxy causal ground truth. It asks a narrower, reproducible question: on the fixed panel, does a model gene graph align with the proxy after controlling measured RNA and graph structure, and is that alignment unusual under both label and target-assignment randomizations? Negative benchmarks of foundation-model interpretability under careful controls have precedent (Ahlmann-Eltze et al., 2025; Boiarsky et al., 2023; Kedzierska et al., 2025); our aim is the audit protocol, software capsule, and reporting discipline rather than a global verdict that scFMs do or do not encode regulation. Failure to support a global capability claim on this panel equally cannot be read as a field-level negative beyond this protocol instance.

The closest prior results show that foundation-model attention recovers co-expression rather than unique regulatory signal (Kendiukhov, 2026b), and that residual-stream features of Geneformer and scGPT likewise encode organized pathway and interaction knowledge with minimal causal regulatory logic (Kendiukhov, 2026a). Our reference differs from transcriptome-derived networks used in those interpretability studies: the proxy edge weights here are built from genomic sequence (motif matches) and chromatin accessibility only, so shared expression statistics alone cannot satisfy the reference through co-expression edges. That construction choice is what makes the audit informative when alignments are small: residual agreement cannot be explained by shared expression edge weights alone, but it also cannot be read as causal TF–target truth. Within one model, we additionally find that two readouts of the same frozen network disagree in sign (Geneformer embedding $+0.0044$ versus attention -0.0036 in brain)—dual-null Support on both sides of zero that is not regulatory recovery and not a model-level capability claim—a per-readout inconsistency that single-readout studies cannot see.

Our contributions are:

1. a reusable fixed-panel audit protocol and protocol-frozen software capsule (panel manifests, null semantics, FDR families, and one-command validation) for scFM gene-graph claims, with biology as the application setting rather than as a claim of recovered regulatory truth;

2. a sequence/accessibility regulatory-potential proxy and 446-TF by 1,200-gene panel that show moderate observed agreement across brain, blood, and a mixed fibroblast reprogramming dataset (motif/proximity stability is not a causal ceiling);

3. a full-confound partial-Spearman statistic that controls co-expression and six structural covariates, with a non-degree specification reported as a prespecified sensitivity analysis;

4. two finite-panel Monte Carlo tests with explicit null semantics, deterministic seed streams, plus-one correction, and eight prespecified FDR families;

5. a multi-readout application of Geneformer, scGPT, scFoundation, and UCE (Theodoris et al., 2023; Cui et al., 2024; Hao et al., 2024; Rosen et al., 2026)—purely transcriptomic FMs, as distinct from knowledge-informed models that inject prior GRN structure at pretraining (Yang et al., 2024; Hu

et al., 2026) and from scBERT (Yang et al., 2022), which is cited as a related transcriptomic FM but is not among the thirteen audited rows—showing small, readout-specific primary alignments—including a sign-inconsistent embedding/attention pair within one model—rather than a uniform positive or negative result; and

6. descriptive per-cell-type, cross-tissue, and axis-aligned injection diagnostics whose scope is kept separate from the pooled inferential families.

These choices are reported as ordinary design rules rather than as a separate boxed checklist. We freeze the TF–target panel; label the reference as a regulatory-potential proxy rather than causal truth; control co-expression and structural covariates; state exactly what each randomization changes; use plus-one Monte Carlo p-values; define FDR families before reading results; and keep exploratory cell-type and sensitivity analyses distinct from the primary test. Relative to scFM interpretability studies, living GRN benchmarks, and scFM task suites (Table 1), scReg-Eval is designed to *complement* rather than compete: it holds motif/ATAC edge weights free of expression statistics, freezes dual-null families, and ships a protocol-frozen audit capsule rather than a leaderboard score. Methods define the panel, proxy, confounds, and nulls; Results report the primary full-confound audit and the prespecified non-degree sensitivity; Discussion and the final figure set bound what the alignments can and cannot claim. Figure 1 states the two-tissue pairing, the ± 2 kb linkage rule, and the frozen panel before the audit heatmaps.

Resource	Expression in reference?	Degree / confound control?	Explicit null semantics?	Multi-readout?	Protocol-frozen capsule?
scReg-Eval (this work)	No (edge weights motif+ATAC)	Yes (coexp + 6 structural)	Yes (two MC nulls + BH families)	Yes	Yes
Kendiukhov et al. scFM interpretability	Yes (co-expression / pathways)	Partial	Varies	Attention-focused	No
Kendiukhov et al. SAE residual stream	Pathway/interaction features	Partial	Varies	Feature-level	No
GeneRNIB living GRN benchmark	Often literature / expression GRNs	Task-dependent	Benchmark scores	Multi-method	Community leaderboard
BioLLM / scFM task suites	Task labels from biology	Task-dependent	Standard ML splits	Multi-task	Code releases

Table 1. Protocol axes versus related evaluation resources. Entries are qualitative design features, not performance ranks. scReg-Eval is a fixed-panel audit capsule; GeneRNIB and BioLLM-style suites score broader tasks against often expression-linked references.

2 METHODS

2.1 Estimand

The target of inference is *fixed-panel unusualness*: on one protocol-frozen TF–target panel, is the partial-Spearman alignment between a frozen foundation-model gene graph and the accessibility/motif proxy unusually large (in absolute value) under (i) a named full or non-degree confound design and (ii) two named randomizations (gene-label permutation and within-TF-row degree-preserving target shuffle), after within-family BH control? Dual-null "Support" answers that panel-local question only. The primary-audit Support column is full-spec / degree-hazard-conditioned by definition. It is not an estimand for causal TF–target effects, not a ranking of foundation models, and not a test of residual regulatory content after every possible confound. Concordance checks against co-expression further bound whether a dual-null row resembles attention-like co-expression geometry; they do not adjudicate residual regulatory truth. Discussion returns to this scope when interpreting Support counts. Four printed uses of "support" are not interchangeable: (i) capitalized Dual-null Support is only the two-null BH conjunction defined below; (ii) lowercase probe "supported" is a sign-flip contrast against co-expression, not Dual-null Support;

(iii) nonzero-edge φ (binary presence of proxy edges) is not an FM gate; (iv) ordinary English "supports a claim" is not a reporting gate. Reporting gates are applied as follows. Dual-null Support is $q_M < 0.05$ and $q_D < 0.05$ under a named confound specification. **Protocol-pass** is the five-gate conjunction of (1) Dual-full Support, (2) FM–co-expression concordance (Spearman > 0; an attention-likeness check), (3) non-degree same sign as the full-spec estimate, (4) multi-readout sign agreement within model family and tissue (single-readout families pass), and (5) full-spec ρ strictly above the tissue co-expression baseline on the same edge set. Dual non-degree Support is reported on every row and is *not* a sixth required gate: under non-degree the row-shuffle still preserves TF out-degree while the statistic does not condition on degree, so the two nulls do not address the same estimand. Multi-readout families on this instance are Geneformer brain (embedding, attention, knockout, and artifact-corrected knockout) and Geneformer PBMC (embedding and attention); every other model$\times$tissue family is a single readout and therefore passes the multi-readout sign gate by definition. Results apply this conjunction to the thirteen available pipeline-complete rows (protocol-pass $0/13$); $7/13$ and $0/13$ are census counts over those rows, not a cross-family multiplicity correction. The family and gate list is printed here.

2.2 Regulatory-potential proxy and fixed panel

The panel contains 1,200 genes, including 446 unique in-range transcription factors, protocol-frozen by a panel manifest in the audit capsule. This is a capsule protocol freeze, not an external preregistration (OSF/AsPredicted) link. Panel construction further requires each gene to have at least one brain snATAC peak in the same ± 2 kb promoter-plus-gene-body window used for linkage, together with RNA-detection and vocabulary filters; the panel is therefore *not* ATAC-orthogonal. Accessible peaks from brain snATAC-seq, paired PBMC multiome, and a chemical reprogramming ATAC dataset (GSE206767, in which day-0 BJ fibroblasts are pooled with day-3 induced neuron-like cells as a single `ALL` accessibility track) are linked to genes by that window rule, which retains 6,609 of 111,743 accessible peaks in the PBMC data (5.91%) and systematically excludes distal enhancer elements. JASPAR2024 CORE vertebrate motifs are scanned with MOODS (Rauluseviciute et al., 2024; Korhonen et al., 2009) at $p < 10^{-5}$ in 500 bp accessible windows, both strands, 554 motifs. Writing accessibility(peak) for the tissue mean accessibility of a window-linked peak and motif-hit(TF, peak) for a MOODS match (1 if any motif of that TF hits the peak, 0 otherwise), the directed proxy weight is

$$G_{\text{proxy}}[\text{TF}, \text{target}] = \sum_{\text{peaks of target}} \text{accessibility}(\text{peak}) \cdot \text{motif-hit}(\text{TF}, \text{peak}).$$

Motif evidence in an accessible peak therefore contributes a directed TF–gene regulatory-potential weight, following the motif-in-accessible-region linkage paradigm used by regulon-inference methods (Aibar et al., 2017). Edge weights are assembled without expression statistics; panel membership and ATAC peak-presence filters remain expression-adjacent inputs, so the proxy is not claimed RNA-structure-independent in every design choice. Accessibility, motif occurrence, and proximity still do not establish direct regulation or causality. We therefore call it a proxy throughout. The two evaluation tissues differ in pairing: the PBMC data are a paired multiome assay (RNA and ATAC from the same cells), whereas the brain analysis combines snATAC-seq with cross-study RNA, so the co-expression control is intrinsically stronger in PBMC than in brain (Fig. 1A).

Brain analyses apply a prespecified neural marker mask to every brain readout; in this panel the mask removes 446 of 534,754 brain edges (0.08%), a small fixed-panel exclusion because only one of the 34 prespecified markers (VCAN, which is not a transcription factor) falls in the fixed panel. PBMC edges are unmasked under the prespecified tissue policy. Before analysis, the driver verifies the manifest, TF identity and order across all caches, matrix dimensions and finiteness, and cache integrity checks.

2.3 Foundation-model and baseline graphs

Gene graphs are read from cached, pipeline-complete outputs over the same manifest. Each readout is a pairwise map G_{FM} on the frozen panel, not a biological assay. Available pooled readouts are Geneformer embedding, attention, and in-silico knockout; scFoundation encoder; UCE encoder (4-layer variant, checkpoint pinned by the pipeline); and scGPT encoder; plus a random-init Geneformer floor. The maps are: embedding = cosine similarity of per-gene contextual embeddings; encoder rows (scFoundation, UCE, scGPT) use that same pairwise embedding map (an alias, not a fourth operator); attention = last-layer gene–gene attention, mean over heads; knockout = an in-silico token-deletion graph readout (not a biological knockout assay), with the artifact-corrected knockout subtracting a random-token position-shift baseline; random-init = a Geneformer-shaped floor scored with the same embedding-cosine operator as the Geneformer embedding row. The primary-audit attention row (Table 3) uses that same *symmetrized* last-layer mean-over-heads map; TF→target and target→TF are orientation sensitivities on the randomization-diagnostic display (marginal Spearman, not the full-spec partial ρ), not additional primary rows. Writing e_j for the contextual embedding of gene j and $e_j^{(-g)}$ for the same embedding after deleting TF g's token, the raw knockout weight is the cell-mean cosine shift

$$G_{\mathrm{KO}}[g,j] = \mathrm{mean}_{\mathrm{cells}}\big(1-\cos(e_j, e_j^{(-g)})\big);$$

the artifact-corrected map subtracts the same shift after deleting a random non-TF token. Coverage differs by tissue because only precomputed, pipeline-complete graphs are included: brain has eight FM/readout rows and PBMC has five. Pairs with a missing cached graph are dropped (no silent impute). The co-expression graph G_{coexp} is the absolute Pearson correlation of library-size CP10k then $\log(1+x)$ RNA on the same non-self TF–gene edge set and the same cell pool as that row (Pratapa et al., 2020). Co-expression is reported as a baseline on the same edge set and nulls, but is not included as an FM row or in FM multiplicity families. Co-expression enters its own test only as the signal, never as a covariate: its design matrix is the full specification minus the co-expression column—an intercept plus the six structural covariates (target peak count, gene length, RNA detection rate, GC content, TF out-degree, target in-degree)—because partialling co-expression out of itself is degenerate by construction. CellPLM is outside scope because its architecture does not expose contextual per-gene states (Wen et al., 2024). Inference settings differ by model and are reported rather than harmonized: PBMC readouts use a deterministic 4,000-cell pool for every model; brain Geneformer embedding and scGPT use all available cells in the preprocessed brain RNA object (those two integers are not separate rows on the coverage display); brain Geneformer attention is capped at 4,000 cells, brain Geneformer knockout uses 300 cells, brain UCE uses 2,000 cells, and brain scFoundation uses 1,500 cells with a 512-token context cap. The coverage display lists those capped or subset rows plus the PBMC 4,000-cell pool. Manifest gene coverage is 1,159/1,200 for PBMC UCE, 1,195/1,200 for brain UCE, 1,199/1,200 for brain scFoundation, 1,168/1,200 for PBMC scFoundation, and 1,200/1,200 otherwise.

2.4 Observed statistic and confound specifications

For each retained TF–target pair, all graph weights are rank transformed. The primary statistic is the Pearson correlation between residualized ranks, equivalent to partial Spearman correlation. The *full* design contains an intercept, ranked co-expression, and standardized target peak count, gene length, RNA detection rate, GC content, TF out-degree, and target in-degree. Target peak count is the count of *all* linked peaks in the gene window (no per-gene cap). GC content is estimated from at most 40 peaks per gene (centered windows on the peak midpoints), so GC understates the extremes of very peak-dense genes while peak count does not. The two degree covariates are computed from the proxy graph itself, so the full specification conditions on functions of the outcome; this is a validity hazard, not a bookkeeping detail, and it is why the *non-degree* design, which omits both degree columns, is reported as a prespecified sensitivity rather than as a footnote. Observed and randomized statistics always use matching column sets. Printed ρ symbols are not interchangeable: ρ_{spec} is this partial Spearman (Pearson of residualized ranks); $\bar{\rho}_{\mathrm{probe}}$ is Spearman of residualized probe vectors (a different operator); ρ_{Mantel} is raw Spearman of unfolded edges; ρ_{resid} is Spearman after subtracting an additive TF-row plus gene-column fit; ρ_{conc} is raw Spearman of an FM graph with co-expression. Writing $r(\cdot)$ for ranks and $x^{\perp C}$ for the residual of x after linear projection onto a named confound matrix C, with spec $\in \{\mathrm{full}, \mathrm{nondeg}\}$ and corr the Pearson correlation,

$$\rho_{\mathrm{spec}} = \mathrm{corr}\big(r(G_{\mathrm{FM}})^{\perp C_{\mathrm{spec}}}, r(G_{\mathrm{proxy}})^{\perp C_{\mathrm{spec}}}\big), \tag{1}$$

where C_{full} is the full design (ranked co-expression mixed with standardized structural columns) and C_{nondeg} omits TF out-degree and target in-degree. The primary-audit, specification-sensitivity, and per-type displays report this statistic.

2.5 Finite-panel randomization inference

The TF panel is treated as fixed; no TF-superpopulation bootstrap (Efron and Tibshirani, 1993) or confidence interval is used. For the gene-label null, one permutation relabels both endpoints of the proxy graph, in the spirit of the Mantel matrix correspondence test (Mantel, 1967). The FM graph, co-expression, and non-degree gene covariates remain fixed. In the full specification, proxy-derived TF out-degree and target in-degree are recomputed after relabeling and then re-standardized in that design.

The second null independently shuffles edge weights within each TF row across non-self targets. This preserves TF out-degree and the row's weight multiset while changing target assignment; target in-degree is recomputed only for the full specification. One proxy randomization per replicate is shared across model rows, with each model scored against the same replicate. Explicit-index tests verify numerical equivalence to separate least-squares calculations, including rank-deficient designs (for example a constant confound column on a row subset).

Both tests use two-sided plus-one correction (Phipson and Smyth, 2010) with $T = \rho_{\text{spec}}$ under the matching column set,

$$p_{\text{MC}} = \frac{1 + \sum_{b=1}^{999} \mathbf{1}\{|T_b| \geq |T_{\text{obs}}|\}}{1000}, \tag{2}$$

with resolution 0.001. The randomization-diagnostic display uses this Monte Carlo p-value. Benjamini–Hochberg correction (Benjamini and Hochberg, 1995) is performed within each of eight families ($m = 8$: tissue (brain/PBMC) $\times$ confound specification (full/non-degree) $\times$ null (gene-label/row-shuffle)). The $7/13$ Dual-null count is a census over the pipeline-complete rows, not a ninth cross-family BH correction. The full specification is the protocol-frozen primary; the non-degree specification is reported alongside it as a prespecified sensitivity because the full specification conditions on proxy-derived degree covariates. Dual-null Support on an FM row is the conjunction of both within-family BH thresholds,

$$\text{Support} \iff q_M < 0.05 \text{ and } q_D < 0.05, \tag{3}$$

with q_M the gene-label (Mantel) BH q and q_D the degree-preserving row-shuffle BH q. The primary-audit display reports Support under this rule; the protocol-pass display uses it as a gate. A sign flip between specifications is

$$\text{sign}(\rho_{\text{full}}) \neq \text{sign}(\rho_{\text{nondeg}}). \tag{4}$$

The specification-sensitivity display marks these sign flips. Dual-null Support is two-sided in $|T|$; protocol-pass additionally requires signed gates (concordance > 0, non-degree same sign, and $\rho >$ baseline). The shared-null Mantel test of $\Delta\rho = \rho_{\text{FM,full}} - \rho_{\text{baseline}} > 0$ ($N = 999$ gene-label draws, one-sided, BH within tissue) is the Results recompute in Table 5. It is not a second primary estimand and is not a protocol-pass gate.

2.6 TF-disjoint supervised probe

The pooled audit is an unsupervised raw-readout test. As a supervised complement, run on paired PBMC multiome data only, we train a ridge probe to predict each TF's proxy-target profile from the FM graph itself. The probe was originally specified over per-gene embeddings; because the pipeline-complete caches store gene-by-gene graphs rather than embeddings, we reformulated it at the pair level without new model inference. Samples are ordered (TF, gene) pairs; features for each family are the edge weight $S_{tf,g}$, its reverse $S_{g,tf}$, and the within-TF rank of the edge. These are scalar per-edge summaries, so the probe tests edge-level linear predictability of the proxy profile, not row-level regulatory programs. Nothing in the design is indexed by TF identity, so a probe trained on 311 held-in TFs (TF-disjoint split, seed 20260730) transfers unchanged to $n_{\text{test}} = 135$ disjoint test TFs. Regularization is chosen by leave-one-out cross-validation over training pairs. Performance is the per-test-TF Spearman correlation between predicted and observed proxy profiles, averaged over test TFs, after residualizing both sides against target peak count, gene length, GC content, and detection rate. Writing C' for that four-column probe design, $\hat{y}_t$ for the ridge prediction on test TF t, and y_t for that TF's observed proxy profile, the probe estimand residualizes first and then takes Spearman (not the rank-then-Pearson operator of ρ_{spec}),

$$\bar{\rho}_{\text{probe}} = \frac{1}{n_{\text{test}}} \sum_t \text{Spearman}\big(\hat{y}_t^{\perp C'}, y_t^{\perp C'}\big). \tag{5}$$

The probe display reports this mean and the paired sign-flip test of per-TF differences versus co-expression. The null permutes gene labels within each test TF; paired FM-versus-co-expression contrasts use sign-flipped per-TF differences. Gene-level degree features are withheld from the primary probe arm because they let the probe reach the proxy through gene-level confounding rather than through the graph's edges.

2.7 Descriptive and pipeline-sensitivity analyses

Per-cell-type rows report observed partial correlations, cell counts, ranges, medians, and sign counts. They have no randomization p-values or FDR adjustment. Cross-tissue proxy reproducibility is also observed-only. On the shared non-self TF–gene edge set, the Mantel rank statistic between two graphs is

$$\rho_{\text{Mantel}}(G,H) = \text{Spearman}\big(\text{vec}(G), \text{vec}(H)\big). \tag{6}$$

The proxy, third-tissue, and appendix Mantel displays report this observed agreement. Let W be the proxy weight matrix on that shared edge set, with i indexing TFs and j genes; μ, a_i, and b_j are the intercept, TF-row, and gene-column terms of an additive fit, and superscripts (A) and (B) are the two tissues in a pair. An additive TF-row-plus-gene-column fit on one tissue's proxy,

$$\begin{aligned} \widehat{W}_{ij} &= \mu + a_i + b_j, \\ f_{\text{add}} &= \frac{\rho_{\text{Mantel}}(\widehat{W}^{(A)}, W^{(B)})}{\rho_{\text{Mantel}}(W^{(A)}, W^{(B)})}, \\ \rho_{\text{resid}} &= \text{Spearman}\big(\text{vec}(W^{(A)} - \widehat{W}^{(A)}), \text{vec}(W^{(B)} - \widehat{W}^{(B)})\big), \end{aligned} \tag{7}$$

is scored against the other tissue; f_{add} is the additive fraction of observed Mantel agreement, not an R^2, and ρ_{resid} is rank agreement after removing each graph's own additive fit. The decomposition display reports these quantities; the appendix construct-lane displays reuse them on construct-lane overlays. Concordance (attention-likeness) is

$$\rho_{\text{conc}} = \text{Spearman}\big(\text{vec}(G_{\text{FM}}), \text{vec}(G_{\text{coexp}})\big) \tag{8}$$

on the same non-self TF–gene edge set (brain 534,308 edges after the marker mask; PBMC 534,754); a readout passes this descriptive gate if $\rho_{\text{conc}} > 0$ (no standard error). The concordance display plots this attention-likeness check; the protocol-pass display uses $\rho_{\text{conc}} > 0$ as a gate. Finally, for each tissue we evaluate the following injected fractions:

$$\alpha \in \{0, .02, .05, .08, .12, .18, .25, .35, .5, .75, 1\}.$$

Writing r_{proxy} for proxy ranks residualized against C_{full} and $z(\cdot)$ for a column z-score (center and scale to unit variance), the injected graph is

$$G_{\text{synth}}(\alpha) = \alpha\, z(r_{\text{proxy}}) + (1-\alpha)\, z(\varepsilon), \tag{9}$$

where ε is independent standard Gaussian noise and the observed statistic ρ_{spec} is repeated 30 times per α. The injection-ladder display reports this ladder. This axis-aligned diagnostic checks implementation response; it is not a power analysis and does not define an MDE, confidence interval, or exclusion bound.

The fixed panel, model/readout availability, inference cell counts, and edge-accounting steps are audited explicitly in the coverage display; blank coverage cells denote missing pipeline-complete graphs, not negative results. The full audit therefore comprises 13 pooled FM/readout rows rather than a balanced model-by-tissue factorial design.

3 RESULTS

The study compares unpaired brain snATAC plus cross-study RNA with paired PBMC multiome, holds a fibroblast ATAC mix as construct-only, and scores a frozen $446 \times 1{,}200$ panel under a promoter-plus-gene-body ± 2 kb linkage rule (Fig. 1).

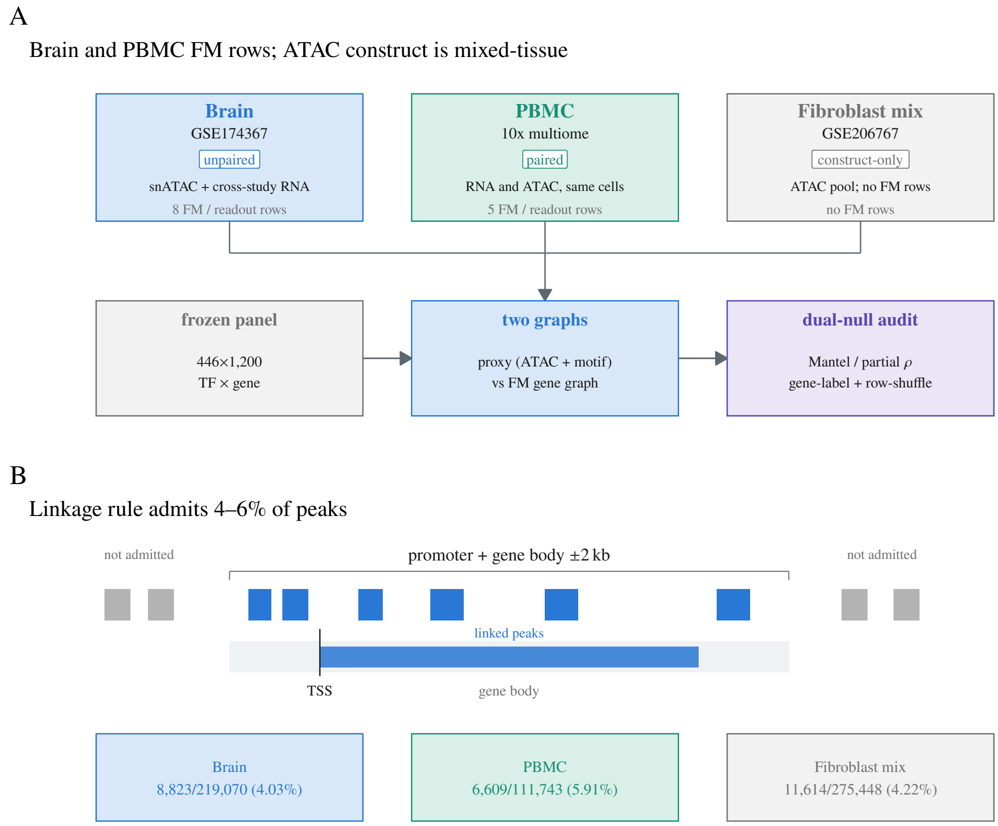

Figure 1. Two-tissue study design and the ± 2 kb linkage rule. (A) Unpaired brain snATAC (GSE174367) plus cross-study RNA versus paired PBMC 10x multiome RNA/ATAC; the fibroblast mix (GSE206767) is construct-only and contributes no foundation-model (FM) rows. All arms freeze the same $446 \times 1{,}200$ panel, then compare an accessibility/motif proxy graph with each FM gene graph under dual-null Mantel/partial ρ. (B) Peaks are admitted only inside a promoter-plus-gene-body ± 2 kb window: 8,823/219,070 (4.03%) in brain, 6,609/111,743 (5.91%) in PBMC, and 11,614/275,448 (4.22%) in the fibroblast mix. Distal and three-dimensional contacts are excluded by this rule.

3.1 The proxy agrees across tissues, but remains a proxy

The accessibility/motif construct shows moderate observed agreement across tissue consensus graphs: brain–PBMC $\rho_{\text{Mantel}} = 0.5249$, brain–fibroblast-mix $\rho_{\text{Mantel}} = 0.5245$, and PBMC–fibroblast-mix $\rho_{\text{Mantel}} = 0.4623$ (Fig. 2; Table 2). The third dataset (GSE206767) is a chemical-reprogramming experiment whose accessibility track pools day-0 fibroblasts and day-3 induced neuron-like cells, so it does not represent any resting tissue. These are descriptive fixed-panel correlations without new randomization inference. They show the construct is stable across ATAC inputs of very different provenance, not a causal ceiling or proof of TF–target truth. Figure 2 also makes the proxy's motif-noise and panel- composition limits explicit. Most of that stability is marginal structure: an additive TF-row-plus-gene-column model fit on one tissue's proxy predicts the other tissue's edges at $\rho = 0.32$–0.41 (69–78% of the observed agreement; Fig. 3), and after removing each tissue's own additive structure the residual agreement is $\rho = 0.41$–0.55. Table 2 reports the same quantities with edge Jaccard and mean per-TF-row Spearman. One pair deviates from that monotone reading: for brain–fibroblast-mix the additive fit is net-subtractive (residual $\rho = 0.55$ above the observed 0.52), so its additive-prediction fraction quantifies similarity in the marginals rather than decomposing the edge-level agreement. The dominant share of the apparent reproducibility is therefore fixed by the motif library and the proximity rule, both of which are tissue-independent by design, rather than by tissue-specific regulatory biology.

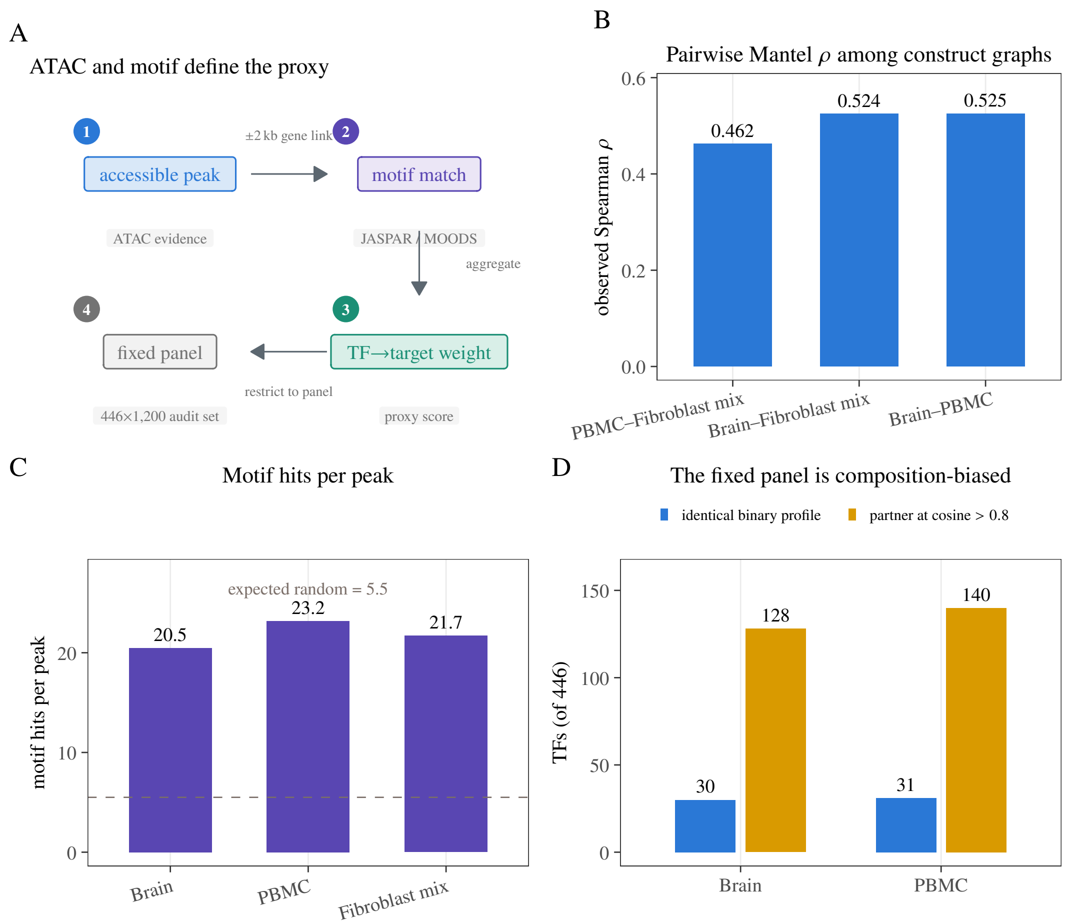

Figure 2. Regulatory-potential proxy, evidence, and panel composition. (A) Construction pipeline: accessible peaks are linked to genes (±2 kb promoter-plus-gene-body), JASPAR/MOODS motif matches contribute directed TF→target weights, and edges are restricted to the frozen 446-TF×1,200-gene panel; edge weights are built without expression statistics (panel and ATAC inputs remain expression-adjacent), and the proxy is not causal ground truth. (B) Observed pairwise Mantel ρ among the three construct graphs (not a claim that the proxy is tissue-reproducible biology). (C) Motif hits per accessible peak; the dashed guide is the expected 5.5 random hits at the 10^{-5} motif threshold (a count, not a share). (D) TF target-profile redundancy: counts with an identical binary profile or a partner at cosine similarity > 0.8, out of 446 TFs. Panel D omits the fibroblast-mix arm (construct-only; no FM composition counts).

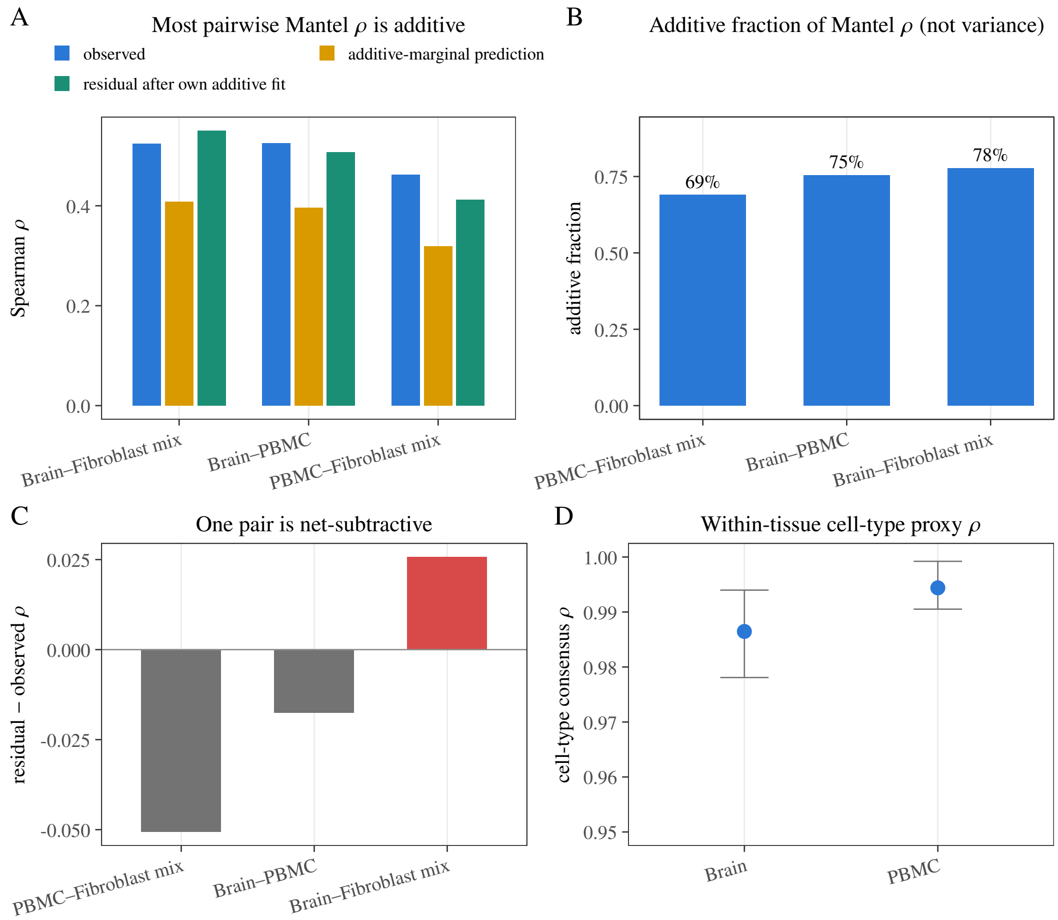

Figure 3. Cross-tissue structure of the regulatory-potential proxy on the frozen $446 \times 1{,}200$ panel (three tissue pairs). (A) Observed Spearman ρ, TF-row-plus-gene-column additive prediction, and residual ρ after removing each tissue's own additive fit. (B) Fraction of observed agreement predicted by those additive marginals (69–78%; an additive fraction of Mantel ρ, not a variance share). (C) Residual minus observed ρ; red = net-subtractive (residual $>$ observed), grey otherwise; brain–fibroblast mix is net-subtractive. (D) Mean pairwise cell-type proxy Spearman ρ within brain and PBMC only; the fibroblast-mix arm is omitted (construct-only; no per-type FM rows). Whiskers are the min–max range (not a confidence interval). These are descriptive fixed-panel quantities without new randomization inference.

Tissue pair	Obs. ρ	Add. frac.	Resid. ρ	φ	Edge Jac.	Mean row ρ	Shared TFs
Brain–PBMC	0.5249	0.7543	0.5073	0.5070	0.416	0.4001	446
Brain–Fibroblast mix	0.5245	0.7771	0.5502	0.5004	0.414	0.4151	446
PBMC–Fibroblast mix	0.4623	0.6895	0.4117	0.4461	0.374	0.3364	446

Table 2. Observed cross-tissue reproducibility of the regulatory-potential proxy on the frozen $446 \times 1{,}200$ panel ($n = 3$ tissue pairs). Columns: observed Spearman ρ_{Mantel}; additive fraction of that agreement from TF-row plus gene-column marginals; residual Spearman ρ_{resid} after each tissue's own additive fit; nonzero-edge φ (phi coefficient of nonzero-edge presence/absence); edge Jaccard; mean per-TF-row Spearman; shared TF count (446). Fixed-panel descriptive quantities; no confidence interval or new randomization inference.

3.2 Primary full-confound audit: six weak positives, one negative, and a baseline that matches them

Table 3 and Fig. 4 report all thirteen pooled FM/readout rows plus the co-expression baseline. Six small positive effects are supported by both randomization tests after within-family BH correction (Support column in Table 3); that full-spec Support count is authoritative only together with prespecified non-degree dual-null Support $0/13$ and protocol-pass $0/13$ (Table 4), not as a standalone recovery claim. In brain, Geneformer embedding has $\rho = 0.00442$ ($q_M = 0.006$, $q_D = 0.003$), UCE encoder has $\rho = 0.00461$ ($q_M = 0.004$, $q_D = 0.003$), and scGPT encoder has $\rho = 0.01241$ ($q_M = 0.004$, $q_D = 0.003$). In paired PBMC, Geneformer embedding has $\rho = 0.00425$ ($q_M = 0.002$, $q_D = 0.003$), scGPT encoder has $\rho = 0.01098$ ($q_M = 0.002$, $q_D = 0.003$), and UCE encoder has $\rho = 0.00593$ ($q_M = 0.002$, $q_D = 0.003$).

Tissue	Readout	Partial ρ	p_M	q_M	p_D	q_D	Support
Brain	Geneformer embed	0.00442	0.003	0.006	< 0.001	0.003	both
Brain	scFoundation encoder	0.00062	0.633	0.633	0.643	0.643	neither
Brain	UCE encoder	0.00461	< 0.001	0.004	< 0.001	0.003	both
Brain	scGPT encoder	0.01241	< 0.001	0.004	< 0.001	0.003	both
Brain	Geneformer attention	-0.00362	0.002	0.005	0.008	0.016	both (neg)
Brain	Random-init floor	0.00187	0.182	0.243	0.175	0.233	neither
Brain	Geneformer KO	-0.00093	0.407	0.465	0.366	0.418	neither
Brain	Artifact-corrected KO	-0.00189	0.100	0.160	0.091	0.146	neither
Brain	Co-expression baseline	0.0028	0.019	–	0.014	–	–
PBMC	Geneformer embed	0.00425	< 0.001	0.002	0.002	0.003	both
PBMC	Geneformer attention	-0.00018	0.866	0.866	0.839	0.839	neither
PBMC	scFoundation encoder	0.00112	0.393	0.491	0.359	0.449	neither
PBMC	scGPT encoder	0.01098	< 0.001	0.002	< 0.001	0.003	both
PBMC	UCE encoder	0.00593	< 0.001	0.002	< 0.001	0.003	both
PBMC	Co-expression baseline	0.0070	< 0.001	–	< 0.001	–	–

Table 3. Primary fixed-panel results under the *full* confound specification only (13 FM/readout rows plus the same-edge co-expression baseline). Partial Spearman ρ is alignment of each graph with the accessibility/motif proxy. M denotes the gene-label null and D the within-TF-row degree-preserving target shuffle ($N = 999$ each); each q is BH-adjusted within its tissue-by-specification-by-null family. Support is dual within-family BH $q_M < 0.05$ and $q_D < 0.05$ ("both"); opposite-sign dual support is labeled "both (neg)"; "neither" otherwise. The co-expression baseline is tested on the same edge set outside the FM families and is marked "–" in Support; its entries are uncorrected p-values (dashes mark where no family correction applies). Dual-null Support is not protocol-pass.

Dual-null Support is only one gate. Table 4 and the protocol-pass display report six gate columns on the same thirteen FM rows (dual-null under full and under non-degree, FM–co-expression concordance (Spearman > 0), non-degree same sign as the full estimate, multi-readout sign agreement within model family and tissue, and full-spec ρ strictly above the tissue co-expression baseline). **Protocol-pass** is the conjunction of five of those gates (Dual non-deg is reported but not required): dual-full, concordance, non-degree same sign, multi-readout sign agreement, and $\rho >$ baseline simultaneously. Under those gates, $0/13$ **rows protocol-pass**—including every dual-null positive—so dual-null Support is not treated as regulatory recovery.

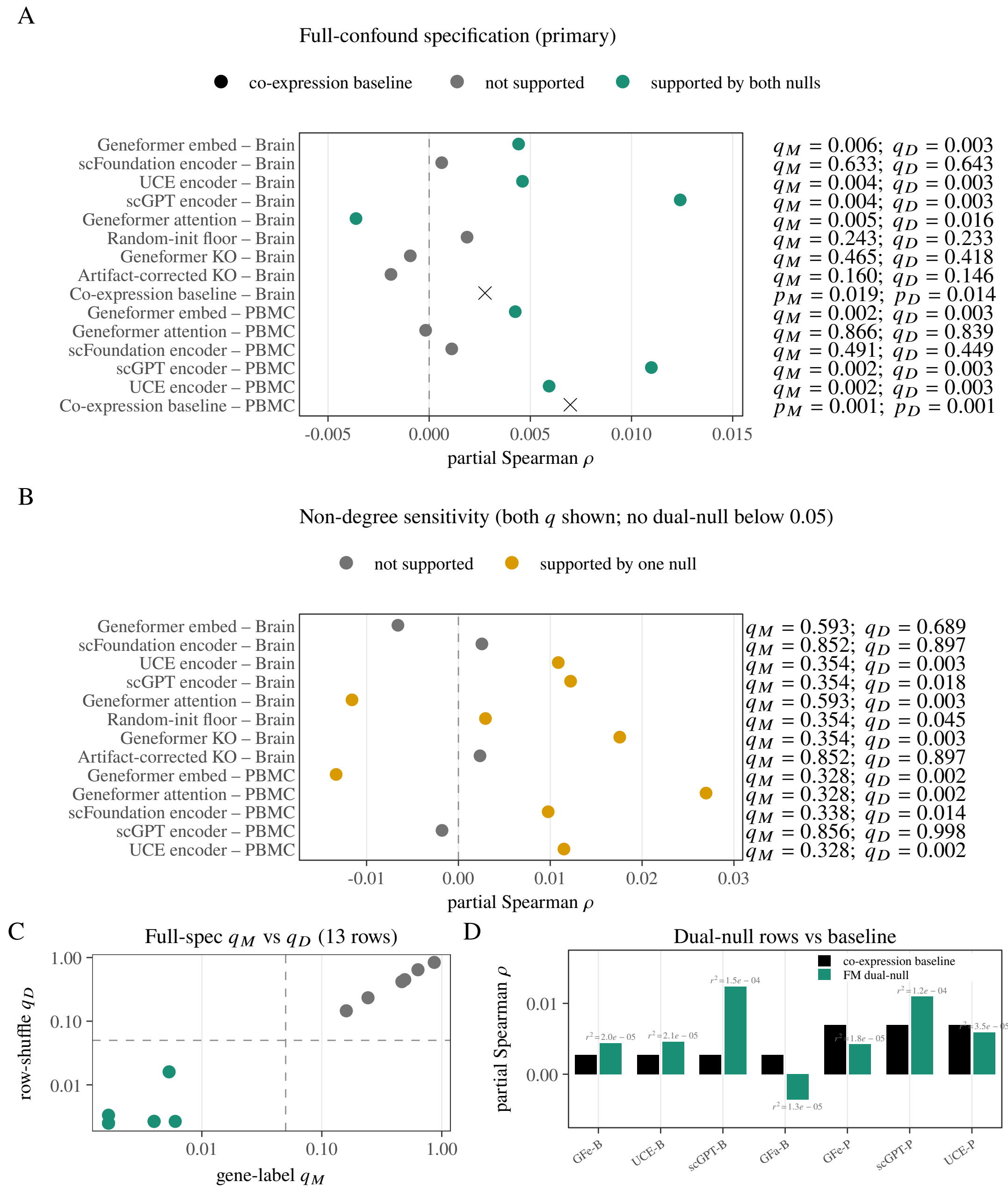

Figure 4. Primary fixed-panel audit of 13 pooled foundation-model (FM) readouts (Geneformer, UCE, scGPT, scFoundation, random-init floor, and Geneformer knockout variants) plus the same-edge co-expression baseline, under both prespecified confound specifications. M is the gene-label (Mantel) null and D the within-TF-row degree-preserving shuffle ($N = 999$ each); FM q values are BH-adjusted within tissue-by-specification-by-null families. Dual-null Support requires $q_M < 0.05$ and $q_D < 0.05$. (A) Full-confound rows; labels are BH q_M, q_D for FM rows and uncorrected p_M, p_D for the baseline (aqua = both nulls, yellow = one null, grey = neither, black = baseline). (B) Non-degree prespecified-sensitivity rows: both q_M and q_D are shown; no row is Dual-null supported (the conjunction $q_M < 0.05$ and $q_D < 0.05$ is empty), while nine rows are D-only ($q_D < 0.05$ with $q_M \geq 0.05$). (C) Full-confound q_M versus q_D; dashed lines at 0.05. (D) Dual-null FM rows versus the tissue co-expression baseline; $r^2 = \rho^2$ on FM bars (tiny effects; several dual-null positives do not exceed baseline). Tick abbreviations: GFe/GFa = Geneformer embedding/attention; B/P = brain/PBMC; UCE and scGPT keep their names. Dual-null Support is not protocol-pass.

Tissue	Readout	Dual full	Dual non-deg	Concord.	ND same sign	Multi-RO sign	$\rho >$ base	Protocol-pass
Brain	Geneformer embed	yes	no	no	no	no	yes	no
Brain	scFoundation encoder	no	no	yes	yes	yes	no	no
Brain	UCE encoder	yes	no	no	yes	yes	yes	no
Brain	scGPT encoder	yes	no	no	yes	yes	yes	no
Brain	Geneformer attention	yes	no	yes	yes	no	no	no
Brain	Random-init floor	no	no	no	yes	yes	no	no
Brain	Geneformer KO	no	no	no	no	no	no	no
Brain	Artifact-corrected KO	no	no	no	no	no	no	no
PBMC	Geneformer embed	yes	no	no	no	no	no	no
PBMC	Geneformer attention	no	no	yes	no	no	no	no
PBMC	scFoundation encoder	no	no	yes	yes	yes	no	no
PBMC	scGPT encoder	yes	no	yes	no	yes	yes	no
PBMC	UCE encoder	yes	no	no	yes	yes	no	no

Table 4. Protocol-pass matrix for the thirteen pooled FM/readout rows (yes = pass, no = fail). Dual full / Dual non-deg: both within-family BH $q_M, q_D < 0.05$ under the named confound specification. Concord.: FM–co-expression Spearman > 0 (attention-likeness). ND same sign: full and non-degree partial ρ share sign. Multi-RO sign: all full-spec readouts of the same model family in that tissue agree in sign (single-readout families pass). $\rho >$ base: full-spec ρ exceeds the tissue co-expression baseline on the same edge set. Protocol-pass is the conjunction of Dual full, Concord., ND same sign, Multi-RO sign, and $\rho >$ base (Dual non-deg is reported but not required beyond same-sign, matching the dual-null failure mode under non-degree). All thirteen rows fail protocol-pass.

One further row is supported by both nulls with the opposite sign: brain Geneformer attention, $\rho = -0.00362$ ($q_M = 0.005$, $q_D = 0.016$), from the same model whose embedding readout is positive in the same tissue. A sign-inconsistent pair of readouts within one model is not what a regulatory-capability account predicts.

The remaining rows are not supported: brain scFoundation ($\rho = 0.00062$), Geneformer knockout (-0.00093) and its artifact-corrected knockout readout (-0.00189; the artifact-corrected row subtracts a random-token position-shift baseline and is not a positive control—the genuine positive control in this audit is the signal-injection diagnostic below), the random-init Geneformer floor (0.00187), and PBMC Geneformer attention (-0.00018) and scFoundation (0.00112).

The co-expression baseline, run through the same nulls on the same edge set (it cannot enter the FM families because partialling co-expression out of itself is degenerate), meets both uncorrected null thresholds in both tissues (outside FM BH families; not FM-family "Support"): brain $\rho = 0.00276$ ($p_M = 0.019$, $p_D = 0.014$) and PBMC $\rho = 0.00697$ ($p_M = 0.001$, $p_D = 0.001$). Two observations follow. First, the baseline reaches the same order of alignment as the supported FM rows. Second, the PBMC Geneformer-embedding positive ($\rho = 0.00425$) is smaller than its own baseline ($\rho = 0.00697$): on that row the model aligns with the proxy less than the confound it is adjusted against. Table 5 reports observed $\Delta\rho = \rho_{\mathrm{FM,full}} - \rho_{\mathrm{baseline}}$ for all 13 FM rows and a shared-null Mantel test of $\Delta > 0$ (N=999 gene-label draws, BH within tissue): only four dual-null full rows have observed $\Delta\rho > 0$ (brain Geneformer embed, UCE, scGPT; PBMC scGPT), and only $2/13$ rows are shared-null significant after BH (brain scGPT; PBMC scGPT), so dual-null Support does not imply multiplicity-controlled superiority to co-expression on the same edge set. Dual-null Support is a reporting gate (both within-family BH $q_M < 0.05$ and $q_D < 0.05$), not an independence Type I rate for the intersection. On the 13 full-spec pairs the two q values are concordant (no mixed one-null full-spec row). The audit records 4 (brain) and 3 (PBMC) dual-null rows; that count is not regulatory recovery.

Tissue	Readout	ρ_{FM}	$\Delta\rho$ vs base	Dual-null	$\Delta\rho > 0$	Dual$\wedge\Delta$>0	q_Δ (shared)	Shared sig
Brain	Geneformer embed	+0.00441	+0.00165	yes	yes	yes	0.379	no
Brain	scFoundation encoder	+0.00056	−0.00220	no	no	no	1.000	no
Brain	UCE encoder	+0.00456	+0.00180	yes	yes	yes	0.379	no
Brain	scGPT encoder	+0.01238	+0.00962	yes	yes	yes	0.008	yes
Brain	Geneformer attention	−0.00353	−0.00629	yes	no	no	1.000	no
Brain	Random-init floor	+0.00188	−0.00088	no	no	no	1.000	no
Brain	Geneformer KO	−0.00110	−0.00386	no	no	no	1.000	no
Brain	Artifact-corrected KO	−0.00197	−0.00473	no	no	no	1.000	no
PBMC	Geneformer embed	+0.00425	−0.00272	yes	no	no	1.000	no
PBMC	Geneformer attention	−0.00018	−0.00715	no	no	no	1.000	no
PBMC	scFoundation encoder	+0.00112	−0.00586	no	no	no	1.000	no
PBMC	scGPT encoder	+0.01098	+0.00401	yes	yes	yes	0.040	yes
PBMC	UCE encoder	+0.00593	−0.00105	yes	no	no	1.000	no

Table 5. FM versus same-edge co-expression baseline under the full confound design. The ρ_{FM} column is a shared-null recompute of the primary partial Spearman printed in Table 3; that table is the only primary ρ. Small differences from the primary fixed-panel ρ_{spec} values reflect that recompute, not a second primary estimand. $\Delta\rho = \rho_{\text{FM}} - \rho_{\text{base}}$ is the observed contrast; q_Δ and Shared sig are one-sided shared-null Mantel tests of $\Delta > 0$ (N=999, BH within tissue). Dual-null is the primary Support gate (both within-family BH $q_M < 0.05$ and $q_D < 0.05$). Dual$\wedge\Delta$>0 counts rows that are dual-null *and* have observed $\Delta\rho > 0$ ($4/7$ dual-null rows; $4/13$ FM rows overall); shared-null significance is $2/13$ (brain/PBMC scGPT only).

Four completed sensitivity checks qualify the fixed-panel interpretation without changing the frozen primary families. Five-fold cross-fitting of the proxy-derived degree covariates preserved the qualitative direction in all 13 pooled rows, with absolute changes from the frozen estimates no larger than 1.1×10^{-5}. Linkage sensitivity preserved the edge-weight structure for TSS–1 kb and TSS–5 kb policies ($\rho = 0.988$ and 0.971 versus the baseline), whereas a promoter-only policy changed the construct substantially ($\rho = 0.119$). In 20 fixed 80% TF subsamples, signs were concordant in all brain and PBMC draws; the descriptive ρ ranges were 0.00271–0.00610 and 0.00115–0.00716, respectively. Detection-based stratification also showed larger alignment for the operational housekeeping proxy than for low-detection TF strata (brain $\rho = 0.00891$ versus 0.00398; PBMC 0.02111 versus 0.00141). These checks report observed stability and construct sensitivity; the TF-subsample check did not recompute valid randomization nulls after subsampling, and the detection split uses an operational label rather than a biological housekeeping annotation.

A separate sequence-opportunity sensitivity applied a five-bin target-opportunity permutation and passed its null-calibration checks, but it is not the frozen primary result: several row-level statistics and q-values differ under that stricter sensitivity design. We therefore report it as a sensitivity audit rather than claiming that the primary headline has been re-tested under that design. External ENCODE pan-biosample TF-occupancy edges on the frozen panel (coverage gate: 204 TFs, 114,467 positive edges) correlate with the pooled motif+ATAC proxy under a lightweight within-TF encode-shuffle null (brain Spearman $\rho = 0.081$, PBMC $\rho = 0.109$; plus-one $p = 0.005$ at $N = 199$; mean per-TF AP ≈ 0.51–0.53 among TFs with ≥ 5 positives). That association is a failed occupancy check: it bounds the proxy as binding-proximal structure and does *not* convert FM dual-null Support into causal TF–target recovery. Our protocol freezes a readout concordance (attention-likeness) check: a co-expression-like readout should correlate positively with co-expression (attention $\approx$ co-expression is well established). This check blocks analogy to attention-GRN recovery stories; it does *not* adjudicate residual regulatory content on its own. Five of the six dual-null positive rows fail concordance: their FM–co-expression Spearman correlations are -0.18 to -0.02 (all negative). PBMC scGPT weakly passes (Spearman $+0.01$) yet still fails protocol-pass (e.g. non-degree same-sign). The readouts that pass concordance are five of thirteen rows: brain scFoundation, brain Geneformer attention, PBMC scGPT ($+0.01$), PBMC scFoundation ($+0.08$), and PBMC Geneformer attention ($+0.40$). Brain Geneformer attention and both scFoundation rows are dual-null negative, so a concordance passer need not be dual-null supported (and conversely). We report the six positives as readout-specific geometry rather than as regulatory signal (Fig. 5). Randomization and knockout diagnostics for the same rows are summarized in Fig. 6; per-TF embedding-norm / cosine anisotropy versus degree is not tested on this instance and is not used as a mechanistic claim.

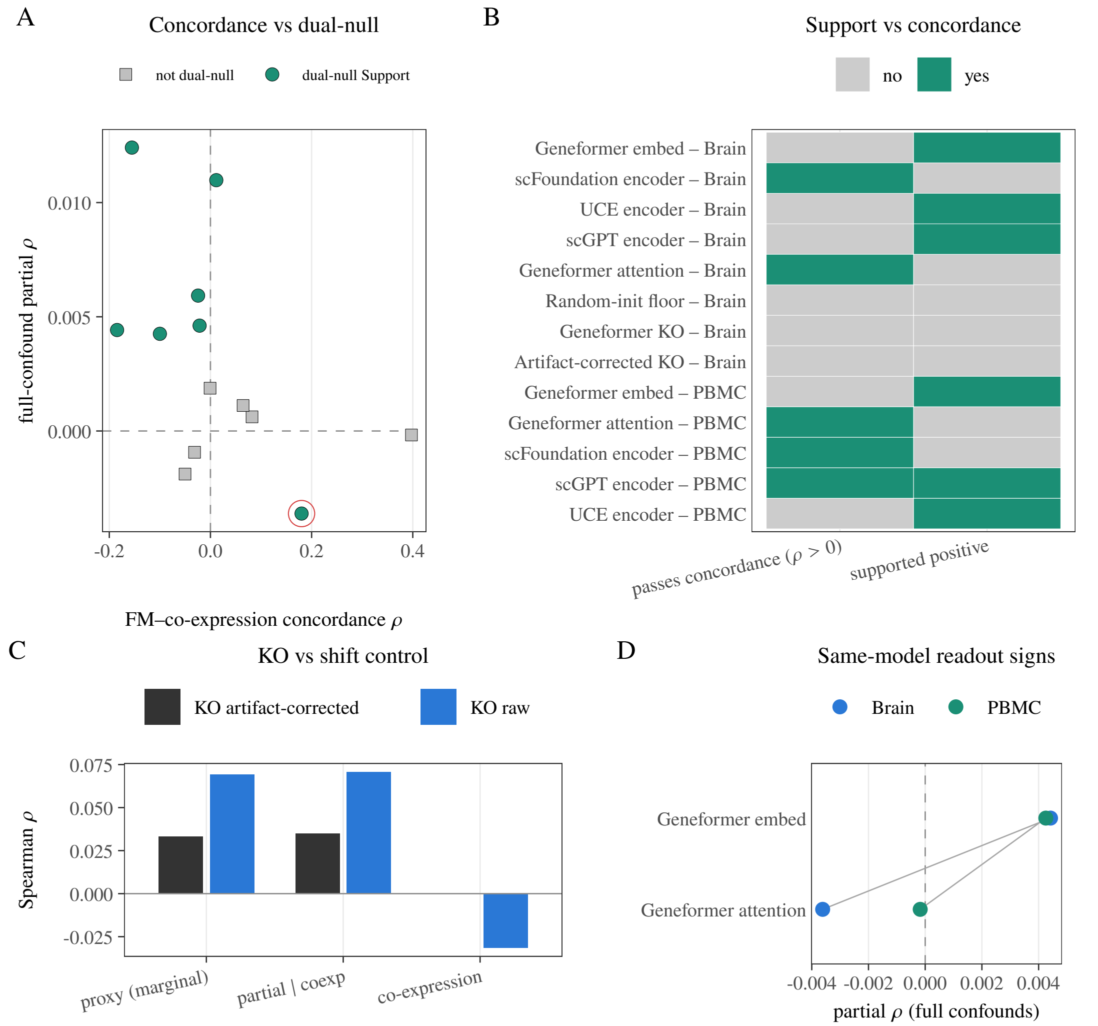

Figure 5. Readout concordance (attention-likeness) and internal consistency for the 13 pooled FM readouts. Concordance is the Spearman correlation of each FM graph with co-expression on the same edge set; it is an attention-likeness check, not a test of residual regulation. Dual-null Support requires both within-family BH q_M and q_D below 0.05. (A) Concordance versus full-confound partial ρ; aqua fill = dual-null Support, grey = not; the open red ring marks brain Geneformer attention (concordance passer, dual-null negative). (B) Positive dual-null Support and positive concordance as separate binary checks (aqua pass, grey fail). (C) Geneformer knockout (KO) raw versus artifact-corrected (position-shift) KO, marginally and after partialling co-expression. (D) Same-model Geneformer embedding versus attention; brain signs are opposite.

3.3 The alignments depend on conditioning on proxy-derived degree covariates

Removing the two degree covariates changes both effect direction and randomization behavior (Fig. 7). No non-degree row is supported by both nulls, and three of the six full-spec positives (brain Geneformer embedding, PBMC Geneformer embedding, PBMC scGPT) are negative before degree conditioning. Because the degree covariates are computed from the proxy graph itself, the full specification conditions on functions of the outcome; the positives therefore appear in exactly the specification that carries this hazard. We report the non-degree specification as a prespecified sensitivity and treat the divergence as a substantive finding, not a footnote. Under non-degree, the audit records **0 dual**, **0 M-only**, and **9 D-only** rejections (row-shuffle $q_D < 0.05$ with gene-label $q_M \geq 0.05$); the gene-label null never rejects alone. Dual-Support is therefore never granted under non-degree. Those D-only rejections must not be read as dual-null Support: the row-shuffle still preserves TF out-degree while the non-degree design does not condition on degree, so the two nulls do not address the same estimand. The non-degree pattern table enumerates that pattern row by row (0 dual; D-only majority).

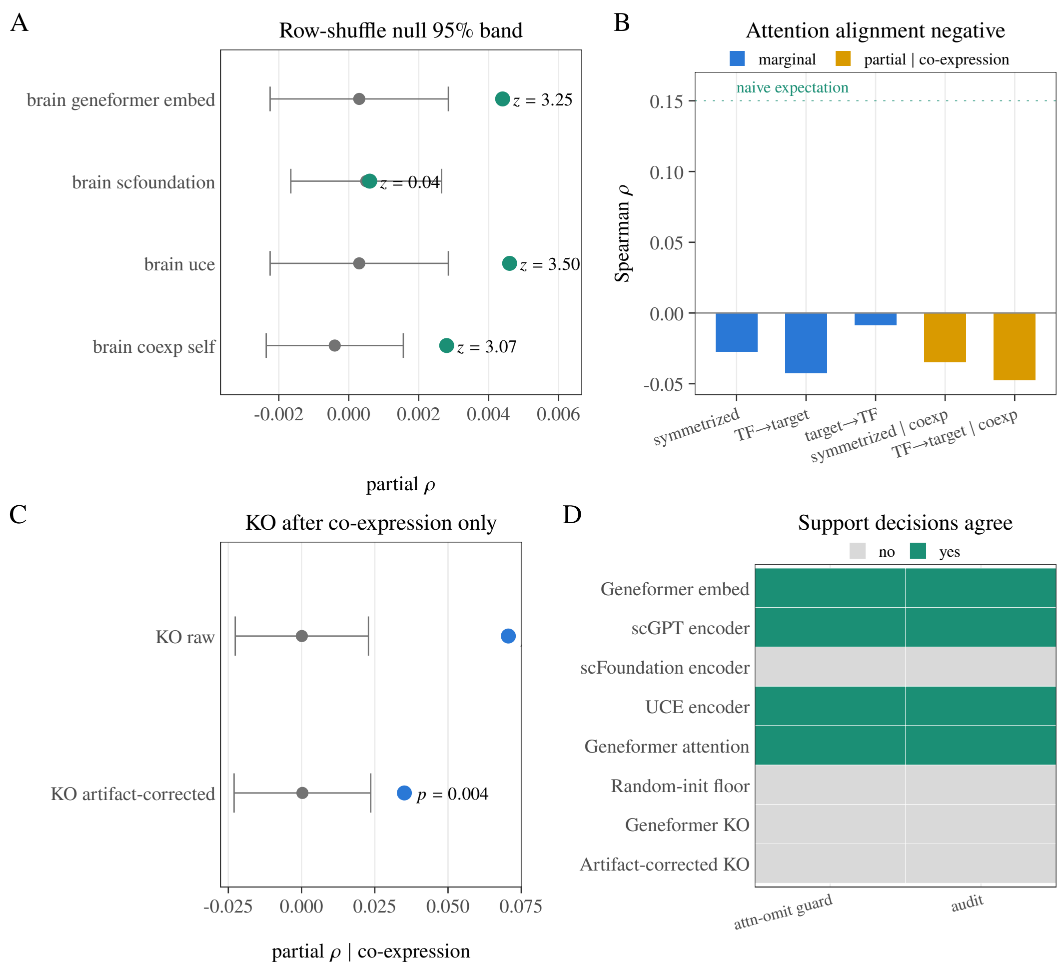

Figure 6. Randomization and readout diagnostics ($N = 999$ permutations). (A) Observed partial ρ (aqua) against the within-TF-row degree-preserving row-shuffle null mean and 95% null band (grey = null mean ± 1.96 SD) for four plotted rows (brain Geneformer embedding, scFoundation, UCE, and the brain co-expression self-test); z is (observed $-$ null mean)/null SD. (B) Brain Geneformer-attention orientation sensitivities: symmetrized, TF$\rightarrow$target, target$\rightarrow$TF, and the two co-expression-partialled variants; the primary-audit attention row uses the symmetrized map. The dotted line is $\rho = 0.15$, a visual reference, not a fitted target. (C) Geneformer knockout (KO) graph readout, raw and artifact-corrected (position-shift), both align with the proxy after partialling co-expression alone (uncorrected $p < 0.001$ and $p = 0.004$, not a BH family); the full confound specification removes that alignment. (D) Dual-null Support decisions agree between the authoritative audit and a parallel attention-omission guard (re-score of the originally omitted attention/floor family on the same eight brain rows; aqua = Dual-null supported).

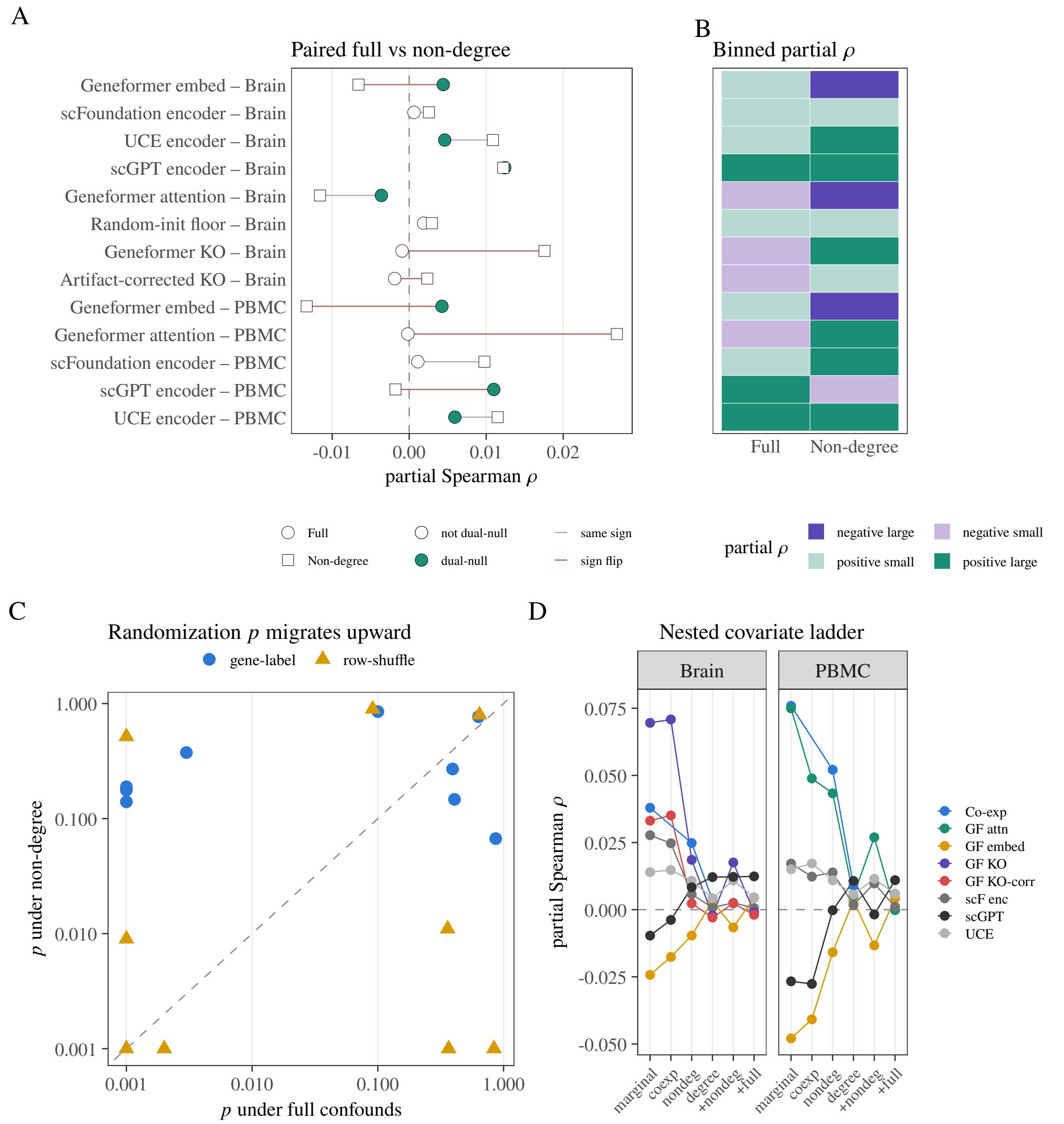

Figure 7. Dependence of the 13 FM alignments on the confound specification. Full includes co-expression plus six structural covariates (including two proxy-derived degree terms); non-degree drops those two degree terms. Dual-null Support is $q_M < 0.05$ and $q_D < 0.05$. (A) Paired full (circles) versus non-degree (squares) partial ρ; aqua fill = dual-null Support (7 under full, 0 under non-degree); red segments = sign flips ($6/13$). (B) Binned partial ρ_{spec} (four magnitude bins at ± 0.005: negative large / negative small / positive small / positive large). This panel is a binned partial-ρ board, not a binary sign map. (C) Full versus non-degree randomization p for the gene-label and row-shuffle nulls. (D) Nested covariate ladder from marginal correlation through co-expression, non-degree, and degree rungs to the full specification; descriptive, not model selection.

3.4 Per-cell-type patterns are descriptive

The proxy itself is near cell-type-invariant on this panel (pairwise consensus $\rho \approx 0.99$), so per-cell-type rows score readouts against a reference with essentially no cell-type contrast; we report them only as descriptive robustness summaries. The full-confound cell-type analysis contains 15 brain (3 readouts $\times$ 5 cell types) and 14 PBMC (2 readouts $\times$ 7 cell types) FM rows per specification (Fig. 8 and Table 6). Brain full-confound estimates range from -0.00396 to 0.00445 with median 0.00258; PBMC estimates range from 0.00222 to 0.00682 with median 0.00413. Those ranges belong with the per-type display, not with the later injection-ladder display. These rows have no randomization p-values or BH correction. They neither support cell-type discoveries nor provide population uncertainty.

Tissue	Confound spec	n_{rows}	$\rho_{\min}$	$\rho_{\max}$	ρ_{median}
Brain	full	15	-0.00396	0.00445	0.00258
Brain	non-degree	15	-0.00986	0.01625	-0.00053
PBMC	full	14	0.00222	0.00682	0.00413
PBMC	non-degree	14	-0.01303	0.05371	0.01502

Table 6. Per-cell-type descriptive ranges of observed partial ρ under full and non-degree confound specifications (brain: 15 exploratory FM$\times$type rows; PBMC: 14). Min/max/median across those rows only; no randomization p-values, BH correction, or population-uncertainty interpretation.

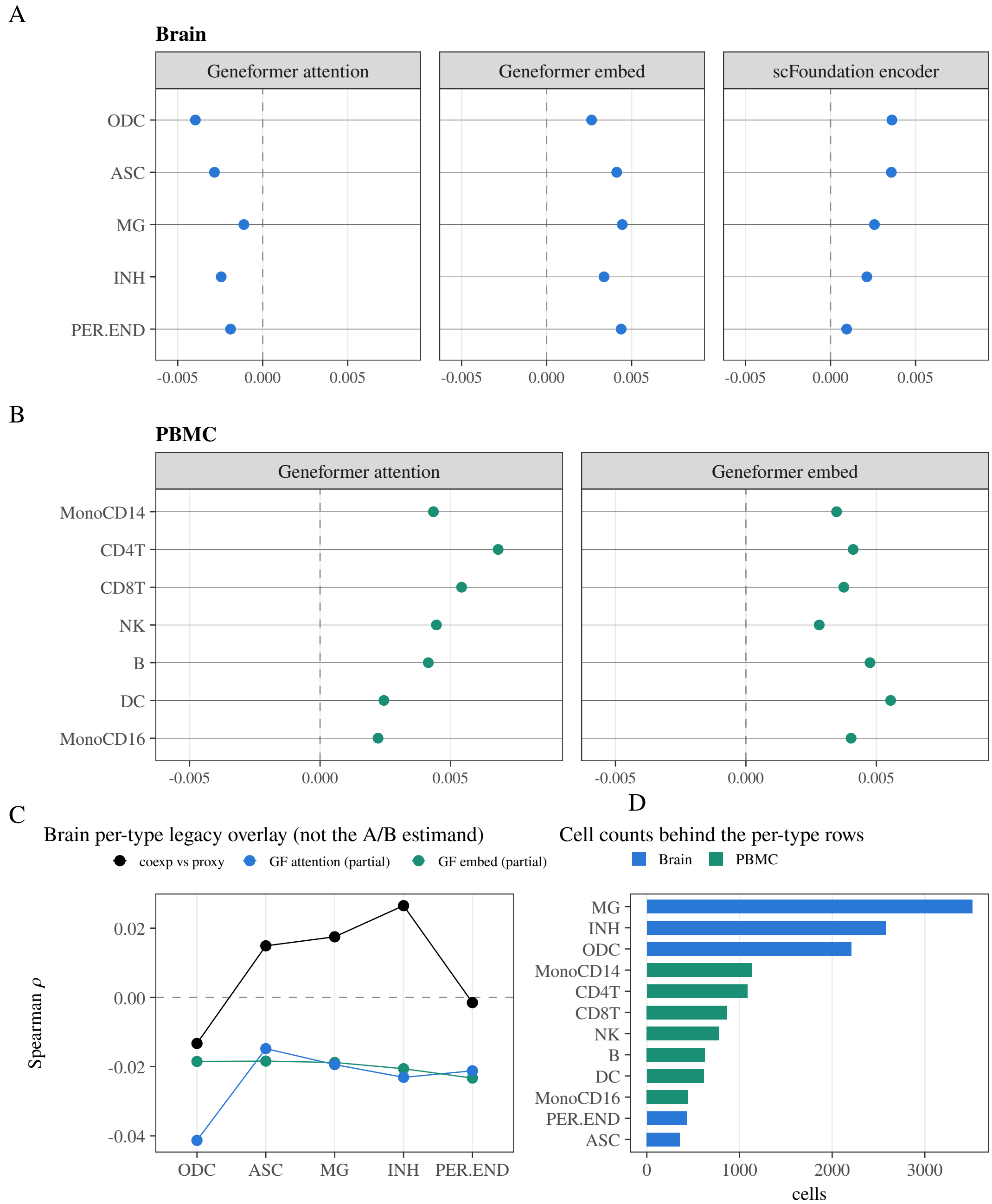

Figure 8. Per-cell-type descriptive robustness summaries of observed partial Spearman ρ under the full confound specification (brain: 15 exploratory FM$\times$type rows; PBMC: 14). (A) Brain rows by readout (x = partial ρ; y = cell type). (B) PBMC rows by readout. The x window is cropped to the Table 6 full-confound ranges (brain -0.00396 to 0.00445; PBMC 0.00222 to 0.00682) with pad so the zero line remains visible. Whiskers in (A–B), where drawn, are bootstrap 95% intervals shown only as descriptive spread; they may clip at the crop and are not randomization p-values or population-uncertainty intervals. (C) Brain per-type legacy overlay: co-expression, Geneformer-embedding, and Geneformer-attention summaries on the same frozen panel (an earlier overlay series, not the A/B estimand and not a second primary family). (D) Cell counts underlying the per-type rows. No multiplicity claims.

Table 7 enumerates the same non-degree pattern row by row.

Tissue	Model	ρ	q_M	q_D	Pattern
Brain	Geneformer embed	-0.00659	0.593	0.689	neither
Brain	scFoundation encoder	$+0.00255$	0.852	0.897	neither
Brain	UCE encoder	$+0.01087$	0.354	0.003	D-only
Brain	scGPT encoder	$+0.01222$	0.354	0.018	D-only
Brain	Geneformer attention	-0.01159	0.593	0.003	D-only
Brain	Random-init floor	$+0.00294$	0.354	0.045	D-only
Brain	Geneformer KO	$+0.01757$	0.354	0.003	D-only
Brain	Artifact-corrected KO	$+0.00235$	0.852	0.897	neither
PBMC	Geneformer embed	-0.01332	0.328	0.002	D-only
PBMC	Geneformer attention	$+0.02696$	0.328	0.002	D-only
PBMC	scFoundation encoder	$+0.00977$	0.338	0.014	D-only
PBMC	scGPT encoder	-0.00178	0.856	0.998	neither
PBMC	UCE encoder	$+0.01149$	0.328	0.002	D-only

Table 7. Non-degree dual-null pattern for all 13 FM rows. The prespecified non-degree sensitivity is the full design minus the two proxy-derived degree covariates. M is the gene-label permutation null and D is the within-TF-row degree-preserving target shuffle ($N = 999$ each); each q is BH-adjusted within the non-degree families. Dual-null Support ($q_M < 0.05$ and $q_D < 0.05$) is $0/13$. Pattern "D-only" is $q_D < 0.05$ with $q_M \geq 0.05$; "neither" is both $q \geq 0.05$. Dual non-deg is reported in the protocol-pass matrix but is not required for protocol-pass.

3.5 The implementation responds monotonically to aligned injection, and the observed effects sit at the bottom of that ladder

The axis-aligned diagnostic rises monotonically from approximately zero at $\alpha = 0$ to mean partial correlations of 0.905 in brain and 0.913 in PBMC at $\alpha = 1$ (Fig. 9). That ladder is a pooled implementation check, not the companion of the per-type range table. The original ladder probes $\alpha \in \{0, .02, .05, .08, .12, .18, .25, .35, .5, .75, 1\}$; the later $\alpha \in \{0.002, 0.005, 0.01\}$ points are subdivided diagnostics used only to locate the observed magnitudes on that curve. They are not direct tests at each row's inferred equivalent and do not constitute a power analysis, MDE, confidence interval, or exclusion bound.

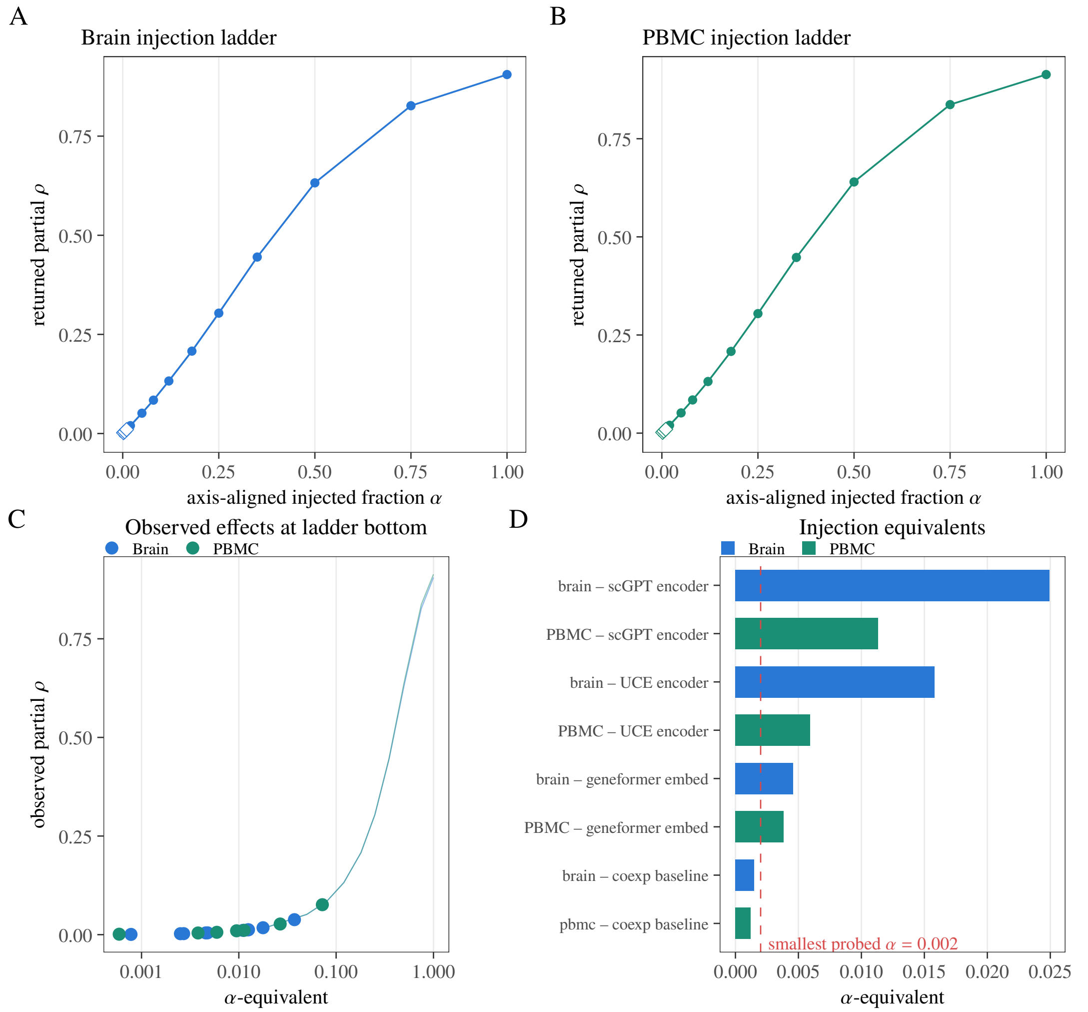

Figure 9. Axis-aligned pipeline-sensitivity diagnostic. α is the injected fraction of the proxy residual mixed with independent Gaussian noise. (A–B) Brain and PBMC ladders: lines are mean returned partial ρ_{spec} of the injected mixture across 30 independent noise replicates, ribbons the replicate range, diamonds the subdivided low-α points. (C) Observed full-spec effects located on those curves. (D) Eight bars: the six dual-null supported *positive* FM rows plus the two tissue co-expression baselines, each as an α-equivalent versus the smallest probed nonzero $\alpha = 0.002$. A bar left of 0.002 means the observed effect is smaller than that first nonzero rung. The negative brain Geneformer-attention row has no positive α-equivalent and is omitted. This is not a confidence interval, power curve, minimum detectable effect, or exclusion analysis.

3.6 A supervised TF-disjoint probe gives mixed family-level results

A linear probe trained to predict each TF's proxy targets from graph features on 311 held-in TFs and tested on 135 disjoint TFs (Fig. 10) gives mixed family-level results after confound adjustment. The probe estimand is supervised TF-disjoint predictability of the proxy profile on **paired PBMC multiome only**; the eight brain rows are not probed. Paired sign-flip testing against co-expression supports the UCE contrast ($q = 0.0325$) and the random-init floor contrast ($q = 0.0300$) when the Benjamini–Hochberg family includes the floor, while Geneformer embedding ($q = 0.5125$), Geneformer attention ($q = 0.65$), and scGPT ($q = 0.255$) are not supported. As a sensitivity check that treats the floor as a control rather than a peer FM, BH recomputed on the $n = 4$ FM contrasts excluding the random-init floor moves UCE to $q = 0.052$ (no longer < 0.05) with the other three FM contrasts remaining non-significant. The sign-flip statistic compares how much each family is over-corrected, not how much it recovers: adjusted test correlations are essentially zero for UCE ($+0.0003$) and the floor ($+0.0023$) but negative for co-expression (-0.0120), so probe "supported" contrasts mean the baseline is over-corrected further below zero, not that UCE carries more residual regulatory signal. The probe's confound set differs from the audit's—it contains neither co-expression nor the proxy-derived degree covariates—so it answers a different question (supervised proxy-profile fit, not raw-readout alignment) and neither confirms nor refutes the pooled rows. In particular, the floor result and the no-floor sensitivity prevent reading the probe as confirmatory evidence that FMs exceed co-expression on this contrast. Table 8 reports the corresponding numeric family-level contrasts (PBMC only; brain omitted), including the no-floor BH column.

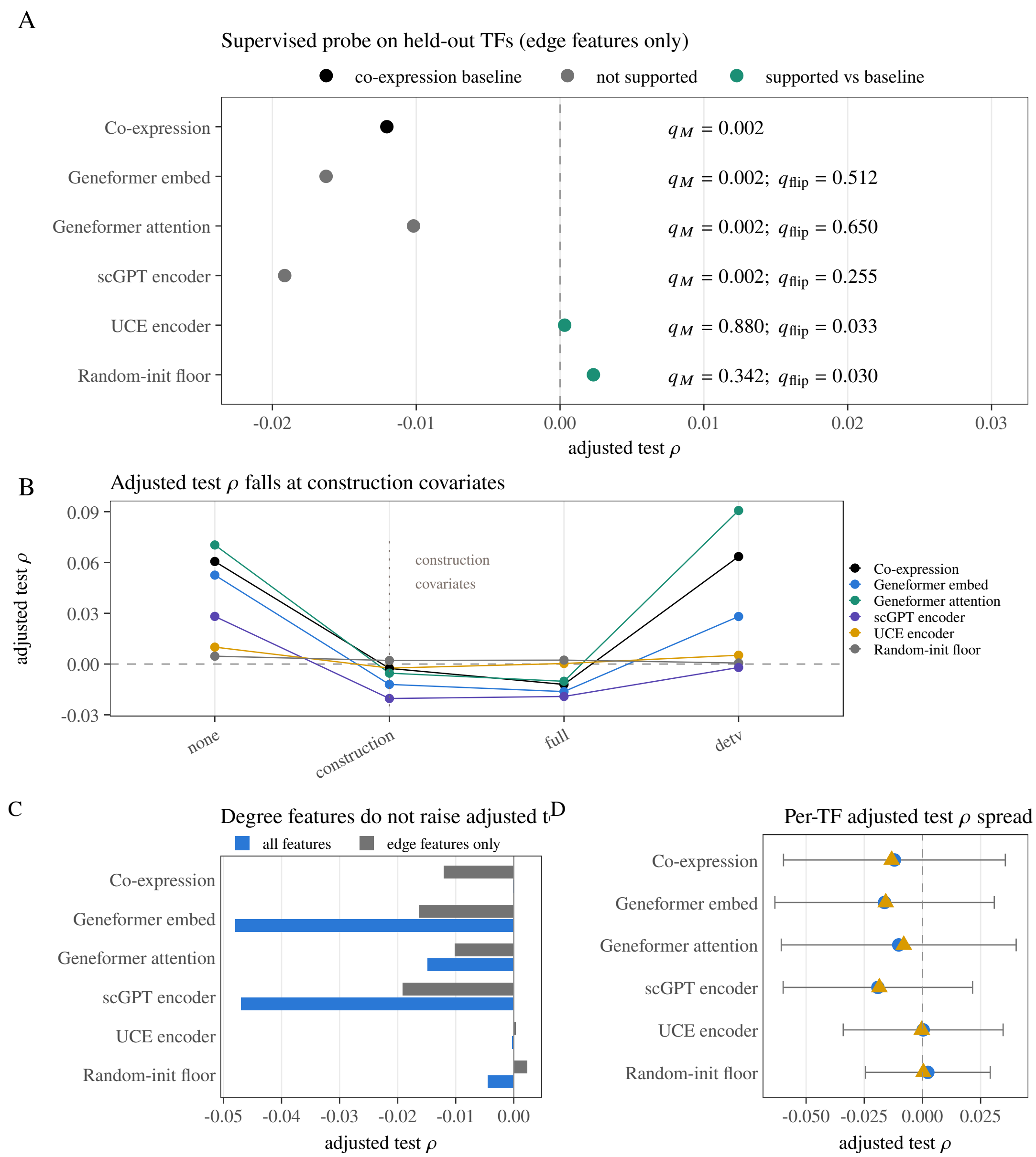

Figure 10. TF-disjoint supervised probe on paired PBMC multiome only (brain rows not probed; train $n = 311$ TFs, test $n = 135$). (A) Confound-adjusted test Spearman per family; gene-label q_M is display-only (not a BH family). For each FM family the paired sign-flip q_{flip} against co-expression is the contrast family. Aqua rows are probe-supported against the baseline at $q_{\mathrm{flip}} < 0.05$ (UCE 0.0325; random-init floor 0.0300) when the floor is in the BH family. (B) Adjusted Spearman across confound subsets (none = empty; construction = peak count + length + GC; full = those plus detection; detection-only = detection rate); the adjusted Spearman falls to near zero at construction covariates. (C) Edge-only primary arm versus the all-feature sensitivity arm. (D) Across-TF mean (circle), median (triangle), and ± 1 SD (whiskers) of adjusted Spearman. The no-floor BH sensitivity ($n = 4$) is Table 8, not redrawn here. Probe contrasts are not residual-regulation claims.

Family	ρ	$\Delta\rho$ vs base	p_{flip}	q_{flip}	q (no floor)
Co-expression	-0.01204	–	< 0.001	0.0015	–
Geneformer embed	-0.01627	-0.00423	0.410	0.5125	0.5467
Geneformer attention	-0.01019	$+0.00185$	0.650	0.6500	0.6500
scGPT encoder	-0.01914	-0.00710	0.153	0.2550	0.3060
UCE encoder	$+0.00032$	$+0.01236$	0.013	0.0325	0.0520
Random-init floor	$+0.00232$	$+0.01436$	0.006	0.0300	–

Table 8. TF-disjoint probe numeric contrasts on paired PBMC multiome only (eight brain rows omitted). A linear probe is trained on 311 TFs and tested on 135 disjoint TFs; $\bar{\rho}_{\text{probe}}$ is the confound-adjusted mean test Spearman. $\Delta\rho$ vs base is versus co-expression. Paired sign-flip q_{flip} is BH on $n = 5$ contrasts (four FM families plus the random-init floor); the no-floor column recomputes BH on the $n = 4$ FM contrasts excluding the floor (UCE moves to $q = 0.052$). Probe "supported" contrasts mean the baseline is over-corrected further below zero, not that the FM recovers residual regulation.

3.7 The construct transfers across three independent ATAC datasets

The proxy was built independently in three datasets (brain GSE174367, PBMC 10x multiome, and the GSE206767 fibroblast reprogramming pool), and Fig. 11 quantifies how much of each dataset the construct actually uses and shares. The ± 2 kb linkage admits only a small fraction of each dataset's peaks: 8,823 of 219,070 in brain (4.03%), 6,609 of 111,743 in PBMC (5.91%), and 11,614 of 275,448 in the fibroblast pool (4.22%)—the proximity-linkage limitation is therefore a three-tissue property, not a PBMC-specific one. On the shared non-self TF–gene edge set (534,754 edges), 38,082 edges carry nonzero weight in all three consensus proxies (24% of the 158,838-edge union), and the TF out-degree marginals are similar across tissues (median 129–171 supported targets per TF)—which is exactly the structure the additive decomposition of Fig. 3 attributes most cross-tissue agreement to. Cross-tissue rank agreement also tracks nonzero-edge agreement (observed ρ_{Mantel} 0.462–0.525 versus presence φ 0.446–0.507), so the construct transfers chiefly at the level of which edges exist, not of their continuous weights. The fibroblast arm carries no foundation-model readouts and enters none of the audit's supported rows; it is a construct-level check, not evidence about model capability.

Independent ATAC overlays on DESCARTES spleen (Domcke et al., 2020), bone-marrow mononuclear cells (BMMC; GSE194122) (Lücken et al., 2021), and regulatory T-cell (Treg) ATAC (GSE211155) (Itahashi et al., 2022) transfer onto the three locked reference graphs at Mantel ρ_{Mantel} 0.391–0.890, with BMMC–PBMC the strongest pair ($\rho_{\text{Mantel}} = 0.890$) and spleen–PBMC the weakest; spleen agreement is largely additive (0.824–0.846) while BMMC–PBMC is not (additive fraction 0.460). Projecting CollecTRI signed transcription-factor–gene interactions (Müller-Dott et al., 2023) onto the same frozen panel keeps 4,592/43,536 raw edges (0.105) and 327/446 TFs, about 19$\times$ sparser than motif-binary nonzero edges (85,809 edges), so the prior cannot replace the accessibility/motif graph. Residual BMMC–PBMC rank match remains high after additive removal ($\rho_{\text{resid}} = 0.950$), yet BMMC covers 1,076/1,200 genes and 341/446 TFs on this panel and the Treg overlay uses filename-sorted metadata; these overlays are outside the 13-row instance and do not add foundation-model audit rows.

Figure 12 applies the five-gate protocol-pass conjunction to the same thirteen rows and bounds what this instance does and does not score.

Coverage, cell counts, and edge accounting for the same frozen panel are collected in Fig. 13.

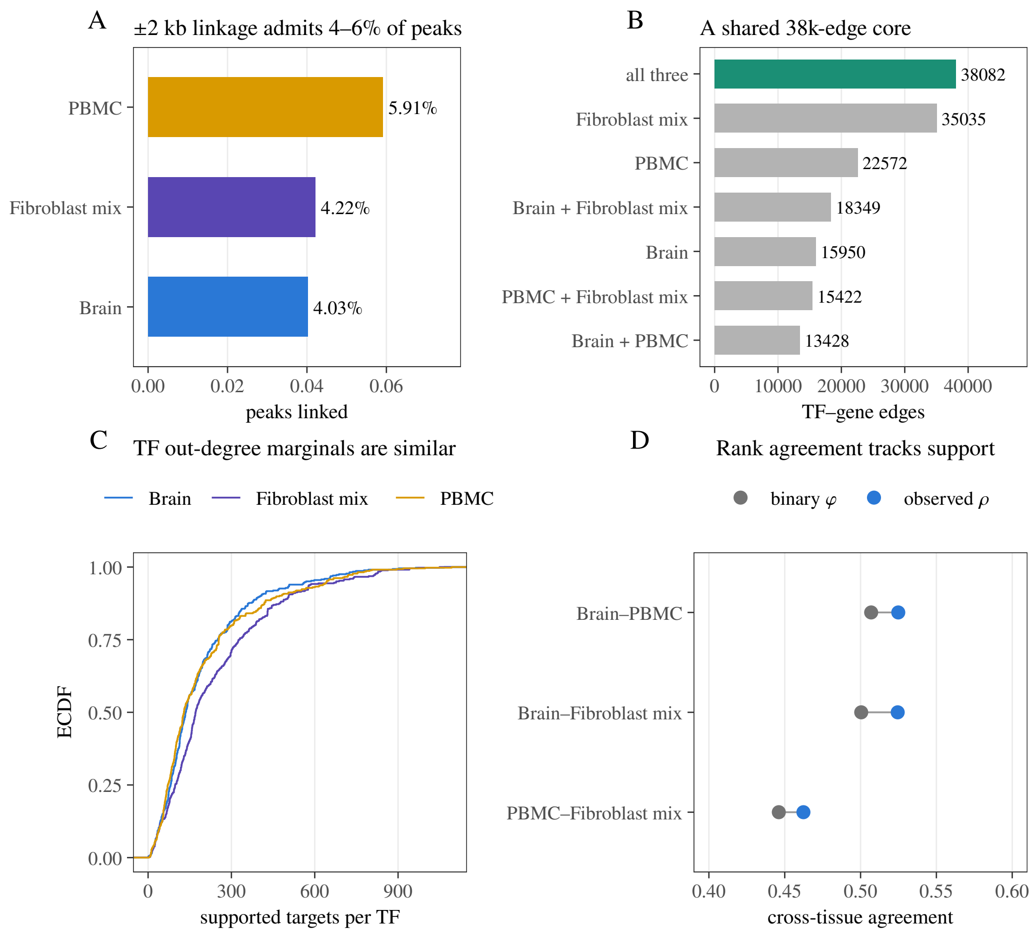

Figure 11. Third-tissue transfer of the proxy construct on brain (GSE174367), PBMC 10x multiome, and the fibroblast mix (GSE206767; construct-only, no FM rows). (A) Fraction of each dataset's peaks admitted by the promoter-plus-gene-body ± 2 kb window (linked over total peaks). (B) Overlap of nonzero non-self TF–gene edges; aqua bar = 38,082-edge core present in all three. (C) Empirical CDF of TF out-degree (nonzero targets per TF; $n_{\mathrm{TF}} = 446$) per tissue. (D) Observed cross-tissue Spearman ρ_{Mantel} versus nonzero-edge φ (phi of nonzero-edge presence/absence) for the three pairs. Descriptive rank/presence agreement only; not a causal TF–target test and not an FM audit.

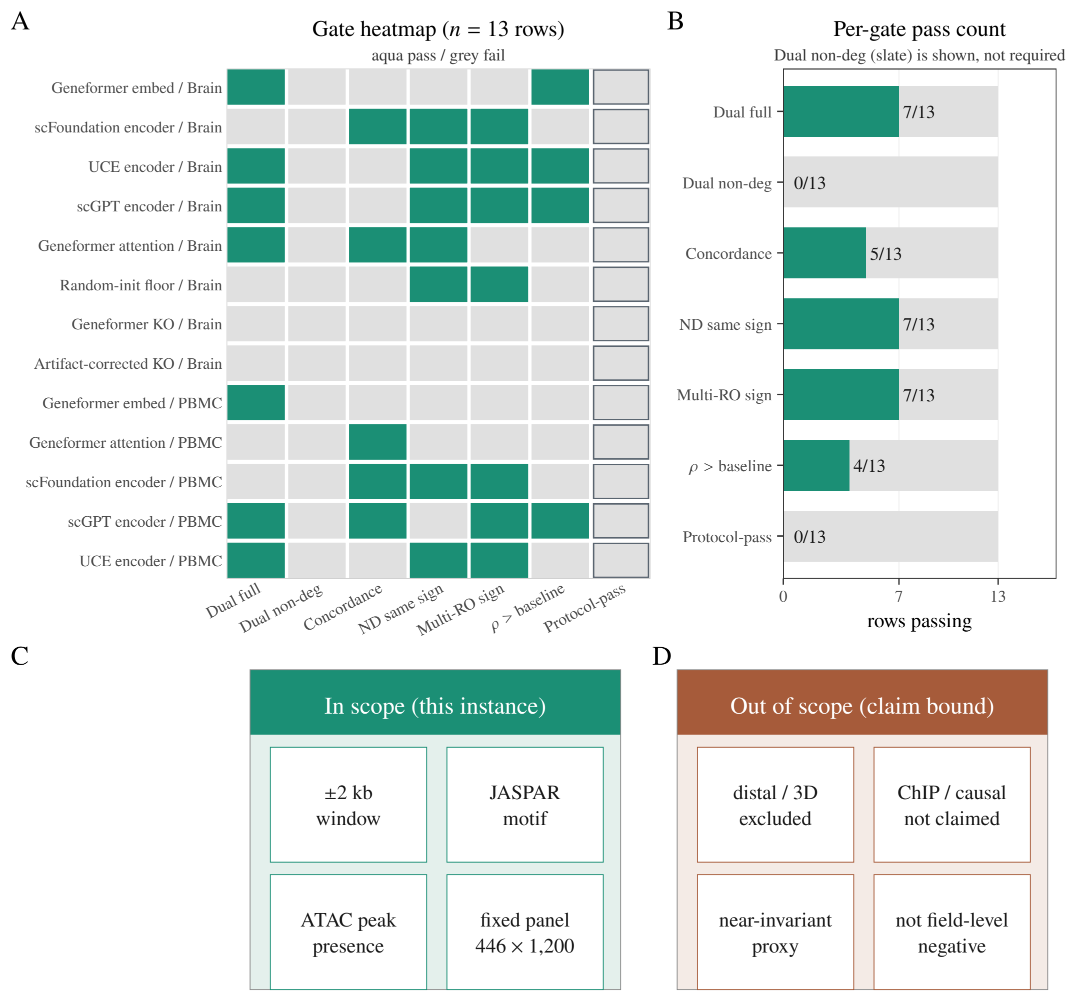

Figure 12. Protocol-pass and instance scope on the thirteen pooled foundation-model (FM)/readout rows. (A) Binary gate heatmap (aqua pass, grey fail). Dual full / Dual non-deg: both within-family BH $q_M, q_D < 0.05$ under the named confound specification. Concordance: FM–co-expression Spearman > 0. ND same sign: full and non-degree partial ρ share sign. Multi-RO sign: all full-spec readouts of the same model family in that tissue agree in sign (single-readout families pass). $\rho >$ baseline: full-spec ρ exceeds the tissue co-expression baseline on the same edge set. Protocol-pass (outlined column) is the conjunction of Dual full, Concordance, ND same sign, Multi-RO sign, and $\rho >$ baseline; Dual non-deg is shown but not required. No row passes ($0/13$). (B) Column sums of panel A: how many of the thirteen rows pass each gate (slate bar = Dual non-deg, reported only). (C) What this instance scores: JASPAR motif matches in ± 2 kb-linked ATAC peaks on the frozen $446 \times 1{,}200$ TF$\times$gene panel. (D) What it does not score: distal/3D structure and ChIP-derived causal TF–target maps; the proxy is cell-type near-invariant on this panel. An external ENCODE occupancy overlay on this panel (brain Spearman $\rho = 0.081$, PBMC $\rho = 0.109$; mean per-TF AP ≈ 0.51–0.53 among TFs with ≥ 5 positives) is a failed occupancy check: it bounds the proxy as binding-proximal structure and does not convert Dual-null Support into causal TF–target recovery. Failure on this instance cannot be read as a field-level negative.

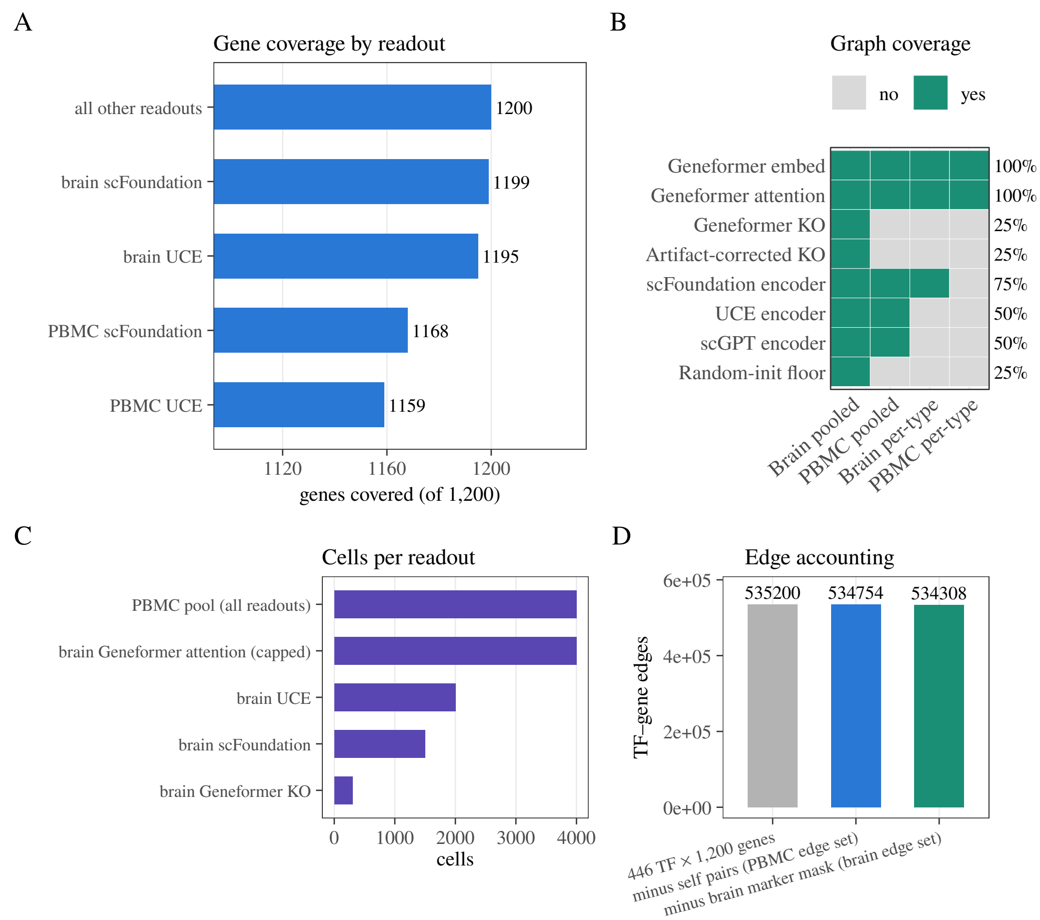

Figure 13. Coverage and fixed-panel quality control on the frozen 446 × 1,200 TF×gene panel. (A) Fraction of each encoder's manifest genes present in the audit gene list (count out of 1,200). (B) Pipeline-complete graph availability by tissue and analysis level (aqua = present, grey = absent); coverage is disclosed rather than assumed balanced. (C) Cell counts used by representative pooled inference runs. (D) Edge accounting from the Cartesian panel through self-pair exclusion (PBMC edge set) and the brain marker mask (brain edge set). This figure is a coverage disclosure, not a proof of balanced design.

4 DISCUSSION

The audit finds six small positive fixed-panel alignments and one negative one, all supported by both randomization nulls under the full confound specification. Three features of that result restrain its interpretation. First, the co-expression baseline meets both uncorrected null thresholds on the same edge set, and a shared-null Mantel test of $\Delta\rho = \rho_{\mathrm{FM}} - \rho_{\mathrm{baseline}} > 0$ supports multiplicity-controlled superiority for only $2/13$ rows (brain and PBMC scGPT; Table 5). Four dual-null positives have observed $\Delta\rho > 0$ versus their tissue baseline; the rest—including PBMC Geneformer embedding—do not. Dual-null Support therefore does not imply separation from the co-expression confound under the frozen $\Delta\rho$ gate. Second, the two Dual-null readouts of the same model in the same tissue (Geneformer embedding $+0.0044$, Geneformer attention -0.0036, both brain) have opposite signs, which is not the signature of a model-level regulatory capability. Third, three of the six positives are negative before conditioning on the proxy's own degree covariates, so they appear only in the specification that conditions on functions of the outcome. A supervised TF-disjoint probe, answering the distinct question of supervised proxy-profile fit on paired PBMC data (brain rows are not probed), finds mixed contrasts: when the BH family includes the random-init floor, UCE and the floor are probe-supported ($q = 0.0325$, $q = 0.0300$); excluding the floor as a peer FM moves UCE to $q = 0.052$. Adjusted test correlations are near zero for UCE/floor and more negative for co-expression, so probe "support" marks relative over-correction of the baseline, not recovered regulatory signal.

The most defensible reading is not that four models encode weak regulation. It is that small, readout-specific, specification-dependent alignments exist on this panel; that protocol-pass is $0/13$; and that under the frozen estimand (Sec. 2.1) dual-null Support does not establish residual regulatory content after named confounds—especially once shared-null $\Delta\rho$ superiority is limited to $2/13$ rows. A further observation points the same way: our own protocol freezes a concordance (attention-likeness) check in which a co-expression-like readout should correlate with co-expression (attention $\approx$ co-expression is well established). Five of the six Dual-null positives fail that check (FM–co-expression Spearman -0.18 to -0.02), while concordance passers include PBMC scGPT ($+0.01$; still fails protocol-pass), scFoundation ($+0.08$), and PBMC Geneformer attention ($+0.40$)—the latter two are dual-null negative—so concordance does not adjudicate residual regulatory content, and dual-null Support without protocol-pass is not recovery. Embedding-cosine anisotropy versus degree remains an untested mechanistic hypothesis on this instance; we do not use it to explain dual-null positives. The pattern is compatible with reports that scFM-derived gene relationships often resemble co-expression (Kendiukhov, 2026b;a), and with zero-shot and biology-driven benchmarks that find scFMs unreliable or only weakly structured relative to simpler baselines (Kedzierska et al., 2025; Wu et al., 2025; Qiu et al., 2025). Failure to support a global regulatory-capability claim on this panel equally cannot support a field-level negative conclusion: the probe is linear and operates on pairwise graph summaries, the proxy is a narrow accessibility/motif construct (Fig. 12C), and other panels, tissues, or GRN-informed models could differ. Models pretrained with explicit GRN inductive biases (Yang et al., 2024; Kalfon et al., 2025; Hu et al., 2026) are outside the present multi-readout audit.

Three statistical caveats apply throughout. First, one proxy randomization per replicate is shared across the rows of a family, so the Monte Carlo p-values within a family are positively correlated; Benjamini–Hochberg remains valid under positive regression dependence (Benjamini and Yekutieli, 2001), but the family-level error rate is then a joint property of the shared randomization—rejections co-occur across rows that share replicates, so the guarantee is weaker than the independent-per-row reading, not stronger. Second, in the non-degree specification the row-shuffle null preserves degree while the statistic does not condition on it, so the two nulls ask different questions there: several non-degree rows are unusual only under the row shuffle (brain Geneformer attention at $q_D = 0.003$ and the random-init floor at $q_D = 0.045$), which we read as evidence that degree adjustment—not regulatory content—drives that null's rejections. Under non-degree, dual-null Support is $0/13$ with 9 D-only rejections (Table 7); we do not treat D-only as Support. Third, replication depth differs by model: scGPT is supported in both tissues but not by the probe; UCE is supported in both tissues but its probe contrast depends on the family boundary; Geneformer is supported only through one of two readouts and its other readout is opposite-signed. We therefore report readout-level, not model-level, support.

Limitations and scope. The reference is a regulatory-potential proxy built from motif and accessibility evidence, and its coverage is narrow by construction. Panel membership further requires ≥ 1 brain

ATAC peak in the linkage window (Sec. 2.2); a no-peak-filter manifest under the same $n = 1200$ cap overlaps the frozen panel in 1167/1200 genes (Jaccard 0.946) and enlarges the pre-cap universe from 12,382 to 12,690 genes. A full dual-null re-audit on that alternate universe was not re-run. The ± 2 kb promoter-plus-gene-body rule retains 6,609 of 111,743 accessible peaks in the PBMC data (5.91%); distal enhancer elements, which carry most cell-type-specific TF occupancy (Thurman et al., 2012), are systematically excluded, and no co-accessibility or 3D-genome linker (for example Cicero (Pliner et al., 2018) or activity-by-contact models (Fulco et al., 2019)) is used. At the MOODS threshold used here ($p < 10^{-5}$, 500 bp accessible windows, both strands, 554 motifs), the expected random hit count is 5.5 per peak while the observed counts are 20.5 (brain), 23.2 (PBMC), and 21.7 (fibroblast mix), so roughly a quarter of motif hits are expected false positives (Wasserman and Sandelin, 2004); no information-content threshold, conservation filter, or PWM-length correction (Stormo, 2000) is applied. The panel is also composition-biased: 30 TFs in brain (31 in PBMC) share an identical binary target profile with at least one other TF (five FOX-family members are one such group), 128 of 446 TFs in brain (140 in PBMC) have a partner at binary cosine > 0.8, and in the reference lineage lists we checked, 7 of 21 neural TFs are present while all 37 housekeeping TFs are, because the panel cap ranks candidates by RNA detection rate. The proxy is near cell-type-invariant on this panel (pairwise $\rho \approx 0.99$), so per-cell-type rows score readouts against a reference with essentially no cell-type contrast. Brain analyses combine unpaired snATAC-seq with cross-study RNA, whereas the PBMC data are paired multiome, so the co-expression control is intrinsically stronger in PBMC than in brain. The full specification conditions on proxy-derived degree covariates; the prespecified non-degree sensitivity avoids that hazard and supports no row under both nulls. The panel is fixed rather than sampled, so p_{MC} describes unusualness under the stated randomizations, not uncertainty over a TF or tissue population; no effect size carries an interval estimate, and with $N \approx 5.3 \times 10^5$ edges, effects of $|\rho| \approx 0.002$ are detectable, so statistical support alone cannot distinguish a small effect from no biologically meaningful effect. With $N = 999$ permutations, 0.001 is the resolution floor of every Monte Carlo p-value, not an exact value; rows with zero null exceedances are reported as $p < 0.001$. The supervised TF-disjoint probe is run on paired PBMC multiome data only; it does not directly test the eight brain rows, whose RNA and ATAC measurements come from unpaired sources. Per-cell-type and cross-tissue analyses are descriptive. The injection diagnostic is axis-aligned and cannot establish power against arbitrary biological alternatives. Fine-tuning, few-shot adaptation, decoder outputs, non-linear probes on raw per-gene embeddings, other gene panels, and direct perturbational validation are not tested. This instance is two FM tissues plus a construct-only ATAC mix on one frozen panel; other RNA collections are outside the instance. ENCODE occupancy is used only as a claim bound (failed occupancy check), not as a scheduled next reference.

The same accessibility/motif construct on independent spleen, BMMC, and Treg ATAC still agrees with the locked references in the Mantel ρ_{Mantel} 0.39–0.89 range, so cross-ATAC transfer is not confined to the three main-text tissues (Fig. A1). A literature prior (CollecTRI) is too sparse on this panel to serve as a substitute reference (Fig. A2), and BMMC covers 1,076/1,200 genes and 341/446 TFs while the Treg overlay uses filename-sorted metadata (Fig. A3). Those panels bound how the proxy moves with new ATAC on this instance; they are not scored against FM graphs.

5 CONCLUSION

scReg-Eval is first a fixed-panel audit protocol and software capsule: frozen panel and nulls, dual-null Support reporting, concordance checks, and baseline comparison on a shared edge set. Applied to thirteen pooled model/readout rows against an accessibility/motif regulatory-potential proxy, it finds six small positive and one negative dual-null alignment under the full confound specification, a co-expression baseline that reaches the same order of alignment, shared-null $\Delta\rho > 0$ significance for only $2/13$ rows (brain and PBMC scGPT; Table 5), protocol-pass $0/13$, sign-inconsistent readouts within the same model, and positives that depend on conditioning on outcome-derived covariates, of which five of six fail concordance (PBMC scGPT weakly passes yet fails protocol-pass). A supervised disjoint-TF probe gives mixed family-level contrasts, with the UCE and random-init floors distinguishing from co-expression but no model-specific regulatory recovery. These application-scene results neither support a global regulatory-capability claim for the audited graphs nor license a field-level negative conclusion beyond this protocol instance. The design complements living GRN benchmarks and standardized scFM task suites (Nourisa et al., 2025; Qiu et al., 2025) by building proxy edge weights without expression statistics while still disclosing panel and ATAC filters as expression-adjacent design choices.

DATA AND CODE AVAILABILITY

Code, fixed-panel manifests, seed metadata, model/readout coverage tables, authoritative audit outputs, and figure sources are available at `https://github.com/PeterPonyu/scfm-reg-audit` and archived at `https://doi.org/10.5281/zenodo.21724336`. The repository and archive support three levels of reproducibility: (i) validation that every published number can be checked without external downloads; (ii) statistical reruns of the fixed-panel audit against the released outputs; and (iii) a full-pipeline recipe with dataset accessions, checkpoint hashes, a locked environment, and ordered commands. Public inputs include GSE174367 brain data (Morabito et al., 2021), the 10x Genomics 10k PBMC multiome (10x Genomics, 2021), and the GSE206767 chemical-reprogramming ATAC pool. Large upstream datasets and model weights are not redistributed; the availability record documents their stable source identifiers and the exact manifests used. GSE206767 is a chemical-reprogramming experiment: it contains day-0 BJ fibroblasts and day-3 induced neuron-like cells, pooled here as one accessibility track because no per-barcode condition metadata is distributed with the matrix (Gene Expression Omnibus, 2023).

ETHICS STATEMENT

This study reuses public, de-identified datasets and performs no new human-subject recruitment or intervention.

COMPETING INTERESTS

The author declares no competing interests.

FUNDING AND AUTHOR CONTRIBUTIONS

This work received no dedicated or external funding.

CRediT author contributions: Zeyu Fu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing—original draft, Writing—review and editing, and Project administration.

DECLARATION OF GENERATIVE AI USE

A large language model accessed through an agentic coding assistant (2026-07 revision cycle) was used as an auxiliary aid for limited engineering edits to three analysis scripts: PBMC UCE co-expression normalization, TF-disjoint probe seed contracts, and layout-independent project-root resolution. No AI system generated scientific data, randomization outputs, or any reported number. The author takes full responsibility for the manuscript and accompanying software. No AI tool is listed as an author. Original code, the prompts used, and the resulting code are not bound inside this PDF.

A ADDITIONAL CONSTRUCT-LANE ATAC OVERLAYS

The thirteen main-text figures report the fixed-panel scReg-Eval audit. The three figures below add construct-lane ATAC from spleen, BMMC, and Treg on the same frozen panel; they are additional data, not a restatement of the main-text proxy figure, not a substitute for Figures 1–13, and they do not add foundation-model audit rows. Spearman ρ is the Mantel rank statistic of Eq. (6) between two TF–gene matrices on $n_{\mathrm{TF}} = 446$ shared transcription factors from the frozen $446 \times 1{,}200$ gene panel. Additive fraction and residual ρ are Eq. (7).

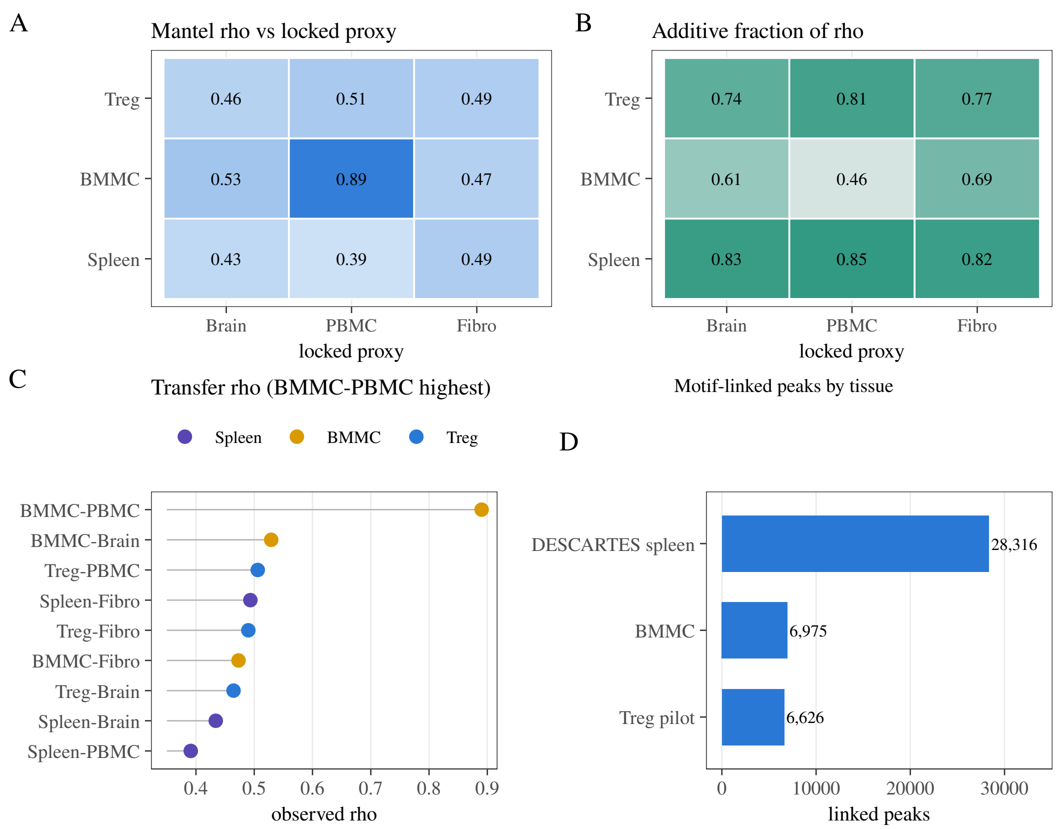

Figure A1. Additional construct-lane ATAC: Mantel transfer of the accessibility/motif transcription-factor–target proxy from spleen, BMMC, and Treg onto the three locked reference graphs. Each overlay graph is a TF $\times$ gene matrix built by scanning JASPAR motifs in accessible peaks and linking hits to genes by a promoter-plus-gene-body rule. Reference graphs use the same construct on adult brain (GSE174367), 10k PBMC multiome, and fibroblast/induced-neuron mix (GSE206767). Overlay tissues are DESCARTES spleen (Domcke et al., 2020), bone-marrow mononuclear cells (BMMC; GSE194122) (Lücken et al., 2021), and regulatory T-cell (Treg) ATAC (GSE211155) (Itahashi et al., 2022). All nine pairs are scored on $n_{\mathrm{TF}} = 446$ shared TFs. **(A)** Observed Spearman ρ. Spleen versus brain/PBMC/fibroblast mix: 0.434, 0.391, 0.493. BMMC versus the same references: 0.529, 0.890, 0.473. Treg versus the same: 0.464, 0.506, 0.490. The BMMC–PBMC cell is the strongest transfer; spleen–PBMC is the weakest. **(B)** Additive fraction of that ρ (TF-row plus gene-column marginals). Spleen: 0.830, 0.846, 0.824. BMMC: 0.608, 0.460, 0.689. Treg: 0.742, 0.805, 0.766. Spleen agreement is largely additive construct machinery; BMMC–PBMC is not. **(C)** The same nine pairs ranked by observed ρ, coloured by overlay tissue. **(D)** Motif-linked peak counts used as construct input: spleen 28,316, BMMC 6,975, Treg 6,626. Peak depth does not monotone-explain transfer ρ. These quantities are descriptive rank agreement. They are not a causal TF–target recovery test and do not add foundation-model audit rows.

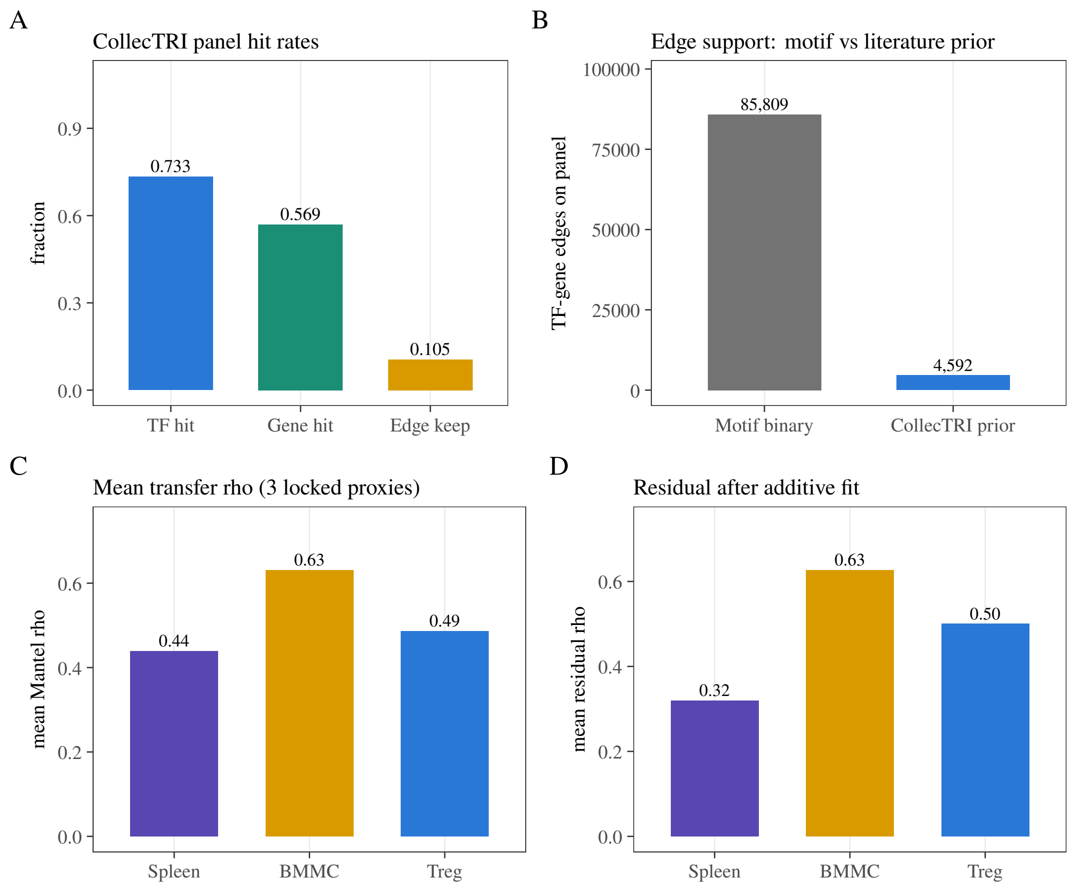

Figure A2. CollecTRI coverage (A–B) and overlay transfer summaries (C–D), not a single CollecTRI estimand across all four panels. CollecTRI signed TF–gene interactions were projected onto the 446 × 1,200 panel (absolute weight; maximum over duplicate edges). Motif-binary nonzero count is the count of nonzero edges in a magnitude-ablated copy of the brain reference graph. **(A)** Hit rates: 327/446 TFs (0.733), 683/1,200 genes (0.569), and 4,592/43,536 raw CollecTRI edges kept (0.105). The prior is TF-rich relative to genes and sparse relative to its own table. **(B)** Edge counts on the panel: motif-binary 85,809 versus CollecTRI 4,592 (about 19× sparser). A prior of this density cannot replace the accessibility/motif graph as a reference. **(C)** Mean Mantel ρ of each overlay against the three reference graphs: spleen 0.44, BMMC 0.63, Treg 0.49. **(D)** Mean residual ρ after additive removal: spleen 0.32, BMMC 0.63, Treg 0.50. BMMC's residual mean matching its transfer mean is the low-additive-fraction fact in another view; spleen's drop 0.44 → 0.32 is the complementary additive fact. CollecTRI is not treated as a causal regulon gold standard.

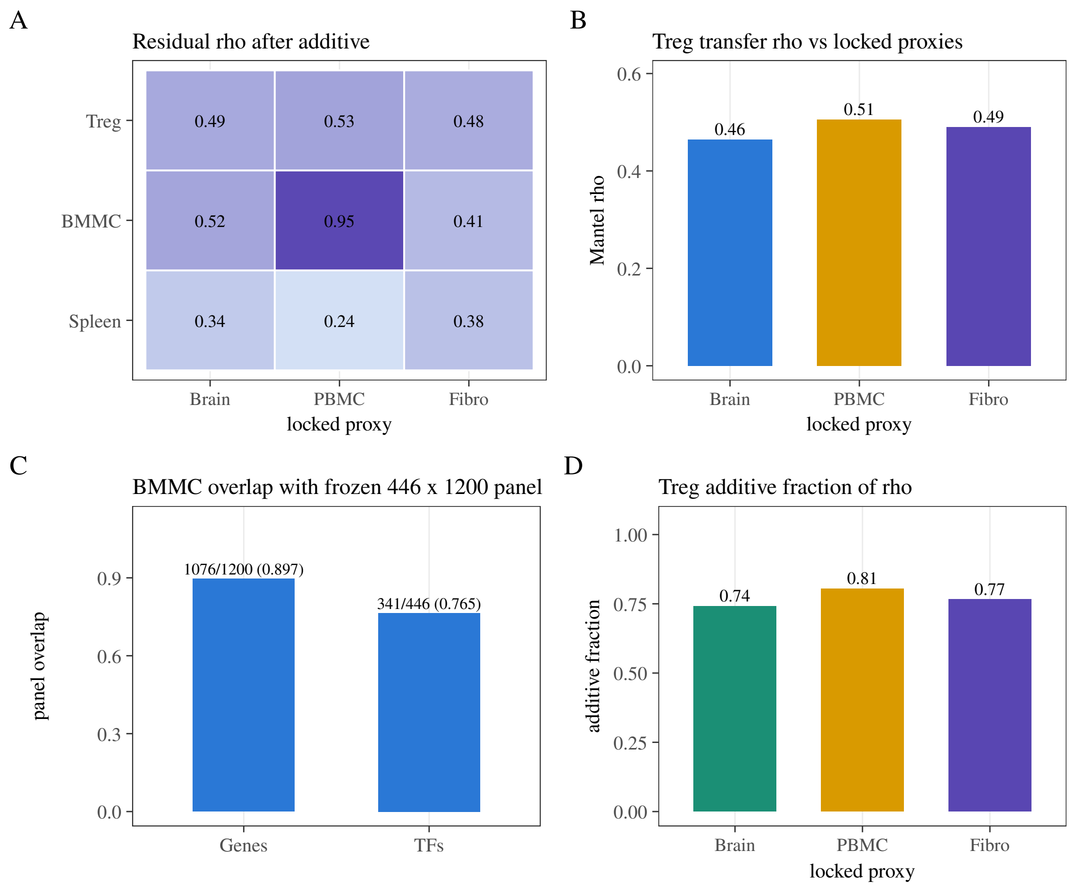

Figure A3. Residual rank match, Treg metadata limits, and BMMC panel overlap on this instance. **(A)** Residual Spearman ρ_{resid} after removing additive TF-row and gene-column fits. Spleen residuals: 0.340, 0.236, 0.383 versus brain, PBMC, and fibroblast mix. BMMC residuals: 0.515, 0.950, 0.414. Treg residuals: 0.494, 0.525, 0.482. Residual ρ_{resid} can exceed observed ρ_{Mantel} when additive structure is net-subtractive (BMMC–PBMC: observed 0.890, residual 0.950). **(B)** Treg Mantel ρ_{Mantel} versus the three reference graphs (0.464, 0.506, 0.490). The Treg population label is taken from filename-sorted metadata rather than from cell-type columns in the peak-matrix object. **(C)** BMMC overlap with the frozen $446 \times 1{,}200$ panel: $1{,}076/1{,}200$ genes (0.897) and $341/446$ TFs (0.765), so BMMC is not a complete panel match on this instance. **(D)** Treg additive fractions (0.742, 0.805, 0.766), showing a more additive profile than BMMC–PBMC. These panels disclose coverage of the overlays that were scored. They do not add foundation-model audit rows.

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